



Princess Margaret Cancer Centre

The AI Prescription: Enhancing Cancer Care with Intelligent Solutions

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Disclosures

- Members of the Team receive royalties from RaySearch Laboratories for license of technology for Machine Learning-based Automated Treatment Planning
- I'm the co-PI of NRG-GU010 which evaluates the use of Darolutamide (Bayer) as a treatment intensification strategy for men with unfavourable intermediate-risk prostate cancer undergoing curative-intent RT.
- I'm a Radiation Oncologist; my views on technology and automation might be biased 😊

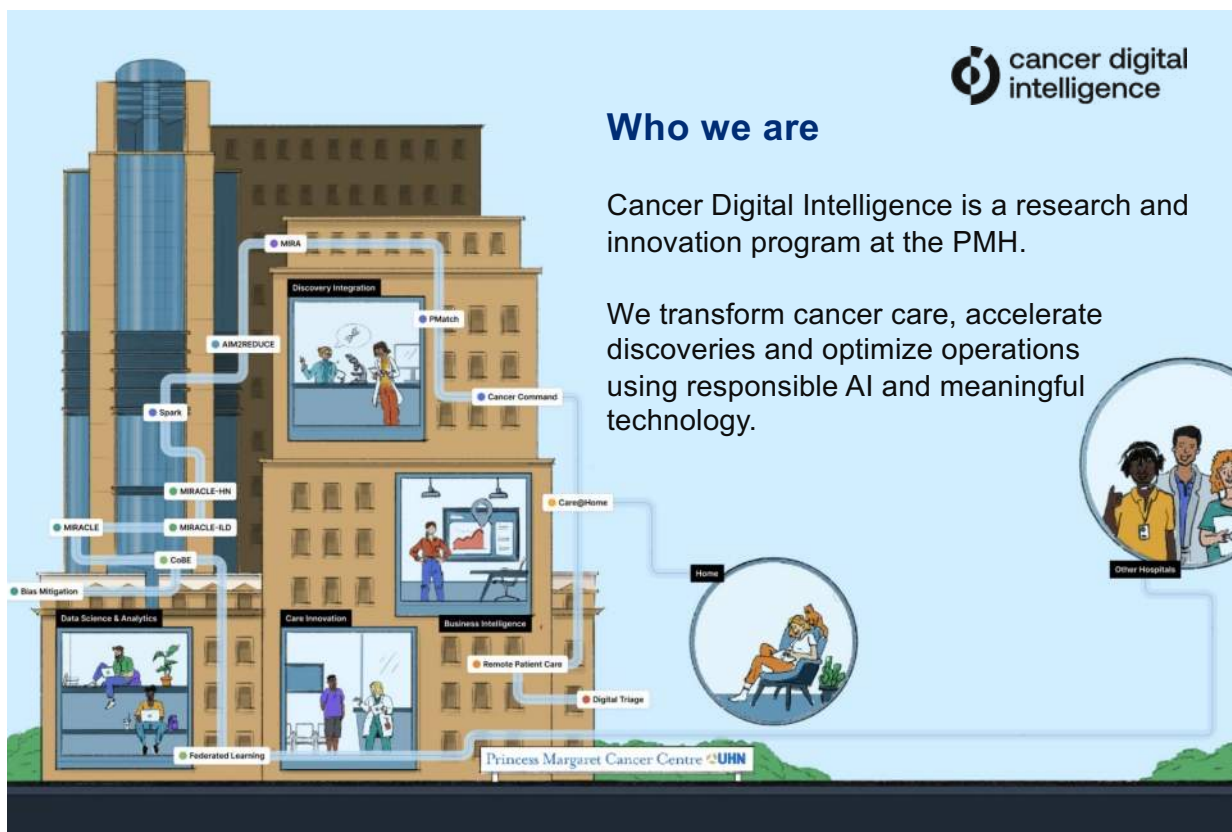
Learning Objectives

- People who attend this presentation would be able to:
 - to understand the current and emerging applications of AI in cancer care, including treatment planning, decision support, and workflow optimization.
 - evaluate the benefits and challenges of AI-driven solutions in oncology and discuss their potential impact on clinical practice and patient outcomes.

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 - evaluate the benefits and challenges of AI-driven solutions in oncology and discuss their potential impact on clinical practice and patient outcomes.
- Ale's Objective: Food for thought & spark some conversations!

Disclaimer: RadOnc-centric .pptx as vehicle to communicate message




Discovery Integration
Bridging the clinic and lab to enable the exchange of data, tools and ideas that accelerate discovery.


Care Innovation
Using digital innovation and service design to improve and streamline the continuum of care.


Business Intelligence
Harmonizing data and analytics to inform strategic decision-making and optimize care delivery.


Data Science and Analytics
Developing robust data infrastructure and advancing novel applications of data science and artificial intelligence (AI) methods.

Disclaimer: RadOnc-centric .pptx as vehicle to communicate message

THE LANCET Oncology

ARTICLES · Volume 24, Issue 8, P936-944, August 2023

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Artificial intelligence-supported screen reading versus standard double reading in the Mammography Screening with Artificial Intelligence trial (MASAI): a clinical safety analysis of a randomised, controlled, non-inferiority, single-blinded, screening accuracy study

Kristina Lång, PhD  ^{a,b}  · Viktoria Josefsson, MD ^{a,b} · Anna-Maria Larsson, PhD ^c · Stefan Larsson, PhD ^d · Charlotte Högberg, MA ^d · Hanna Sartor, PhD ^{a,b} · Prof Solveig Hofvind, PhD ^{e,f} · Ingvar Andersson, PhD ^{a,b} · Aldana Rosso, PhD ^g [Show less](#)

 Check for updates

original reports Multi-Institutional Validation of Deep Learning for Pretreatment Identification of Extranodal Extension in Head and Neck Squamous Cell Carcinoma

Benjamin H. Kann, MD¹; Daniel F. Hicks, MD²; Sam Payabvash, MD³; Amit Mahajan, MD⁴; Justin Du, BS⁵; Vishal Gupta, MD⁶; Henry S. Park, MD, MPH⁶; James B. Yu, MD⁶; Wendell G. Yarbrough, MD, MMHC⁶; Barbara A. Burtress, MD⁶; Zain A. Husain, MD⁷; and Sanjay Aneja, MD⁸

Article

Automated real-world data integration improves cancer outcome prediction

<https://doi.org/10.1038/s41586-024-08167-5>

Received: 9 January 2024

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Published online: 06 November 2024

Open access

 Check for updates

Justin Jee^{1*}, Christopher Fong^{1*}, Karl Pichotta^{1*}, Thanh Ngoc Tran^{1*}, Anisha Luthra^{1*}, Michele Waters¹, Chenlian Fu¹, Mirella Altos¹, Si-Yang Liu¹, Steven B. Maron^{1,2}, Mehraj Ahmed¹, Susie Kim¹, Mono Pirun¹, Walid K. Chatila¹, Ino de Bruijn¹, Arfath Pasha¹, Ritika Kundra¹, Benjamin Gross¹, Brooke Mastrogilacomo¹, Tyler J. Aprati¹, David Liu¹, JianJiong Gao¹, Marzia Capelletti¹, Kelly Pekala¹, Lisa Loudon¹, Maria Perry¹, Chaitanya Bandlamudi¹, Mark Donoghue¹, Baby Anusha Satravada¹, Axel Martin¹, Ronglai Shen¹, Yuan Chen¹, A. Rose Brannon¹, Jason Chang¹, Lior Braunstein^{1,2}, Anyi Li¹, Anton Safonov¹, Aaron Stonestrom¹, Pablo Sanchez-Vela¹, Clare Wilhelm¹, Mark Robson^{1,2}, Howard Scher^{1,2}, Marc Ladanyi¹, Jorge S. Reis-Filho¹, David B. Solit¹, David R. Jones¹, Daniel Gomez¹, Helena Yu¹, Debyani Chakravarty¹, Rona Yaeger^{1,4}, Wassim Abida^{1,4}, Wungki Park^{1,4}, Eileen M. O'Reilly^{1,4}, Julio Garcia-Aguilar^{1,4}, Nicholas Socci¹, Francisco Sanchez-Vega¹, Jian Carrot-Zhang¹, Peter D. Stetson¹, Ross Levine^{1,4}, Charles M. Rudin^{1,4}, Michael F. Berger¹, Sohrab P. Shah¹, Deborah Schrag^{1,4}, Pedram Razavi^{1,4}, Kenneth L. Kehl¹, Bob T. Li^{1,4}, Gregory J. Riely^{1,4}, Nikolaus Schultz^{1,5} & MSK Cancer Data Science Initiative Group

nature cancer



Article

<https://doi.org/10.1038/s43018-025-00991-6>

Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

Received: 13 August 2024

Accepted: 29 April 2025

Published online: 06 June 2025

 Check for updates

Dyke Ferber^{1,2}, Omar S. M. El Nahhas², Georg Wölflein³, Isabella C. Wiest^{2,4}, Jan Clusmann^{2,5}, Marie-Elisabeth Leßmann^{2,6}, Sebastian Foersch⁷, Jacqueline Lammert^{4,8,10,11}, Maximilian Tschochohel¹², Dirk Jäger¹, Manuel Salto-Tellez¹³, Nikolaus Schultz^{1,4}, Daniel Truhn¹⁰ & Jakob Nikolas Kather^{1,2,12}

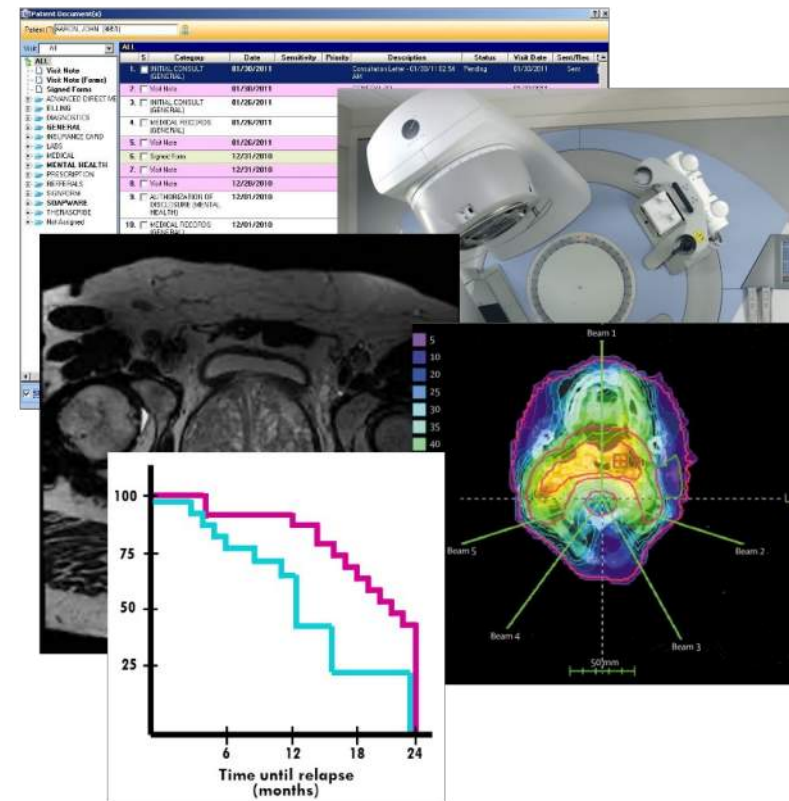
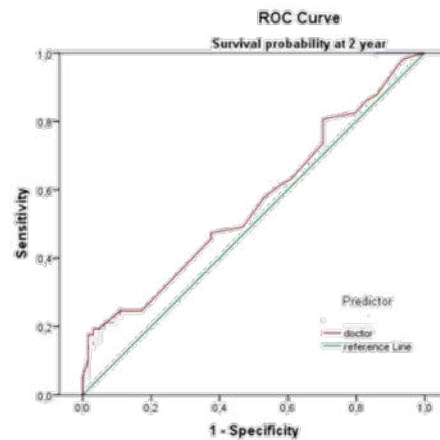
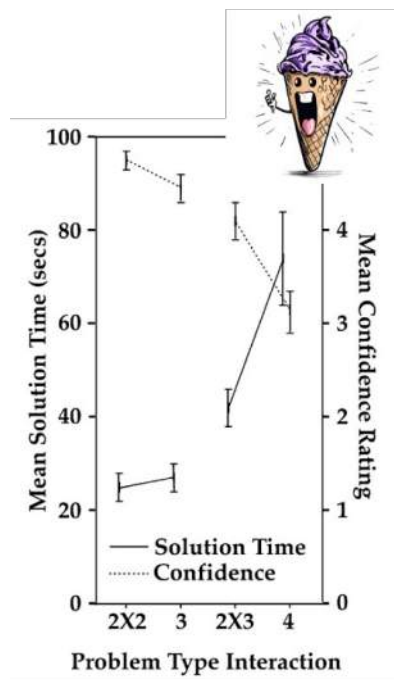
It all started not to long ago!!!

“... a 2-month, 10-man study of artificial intelligence carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that ***every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.*** An attempt will be made to find how to make machines ***use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves.***”



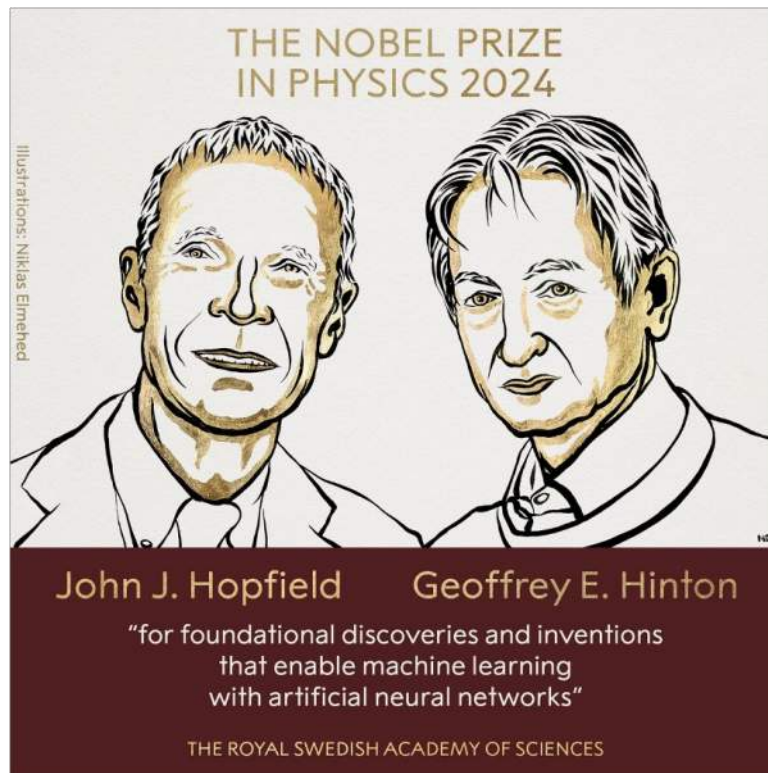
1956 - The Dartmouth Summer Research Project on Artificial Intelligence

Let's agree: "humans are rather bad at making multifactorial decisions"



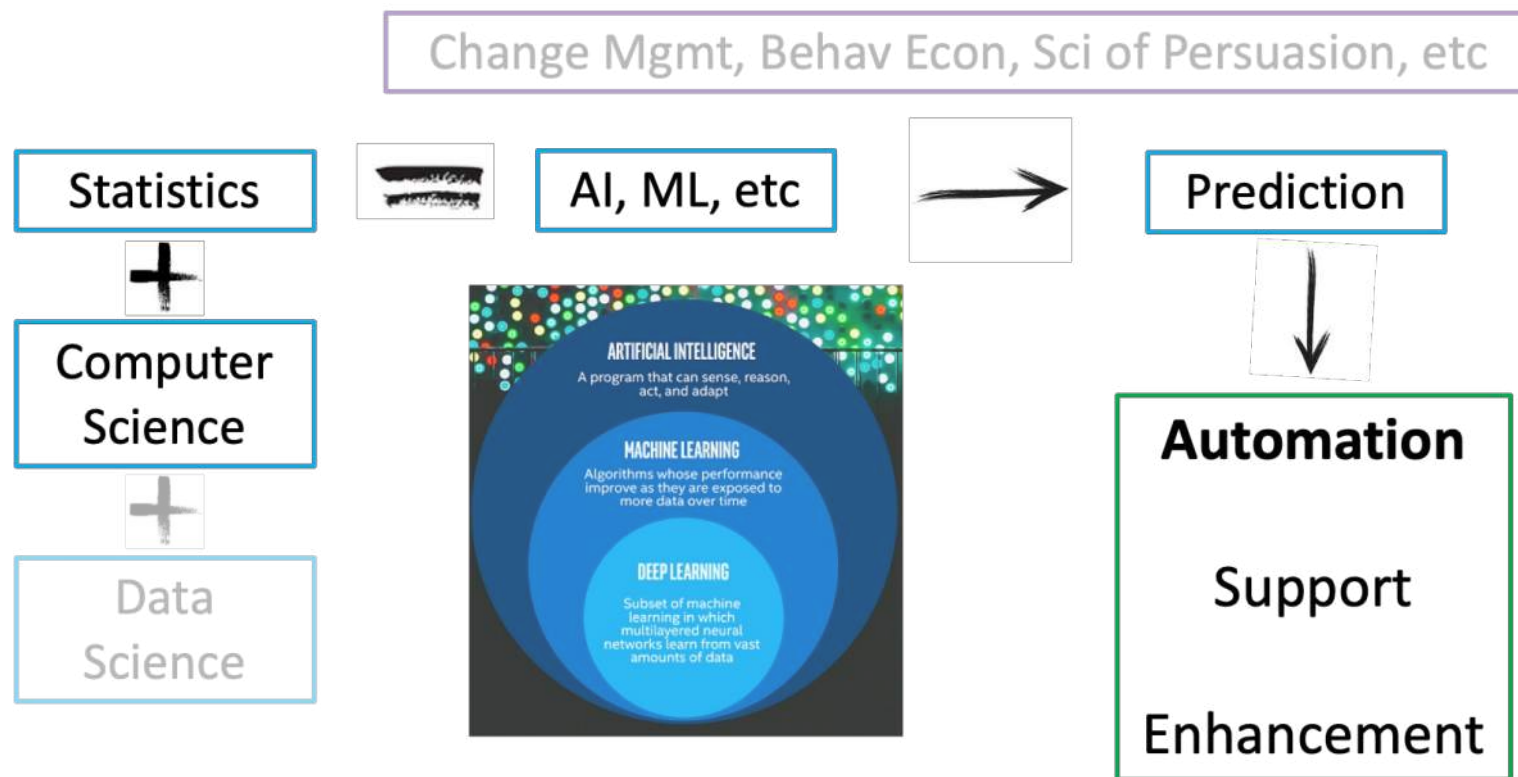
Halford et al. Psych Sci. 2005; Dekker et al (ESTRO 2014)

Let's agree: "AI constitutes a (new) reality"

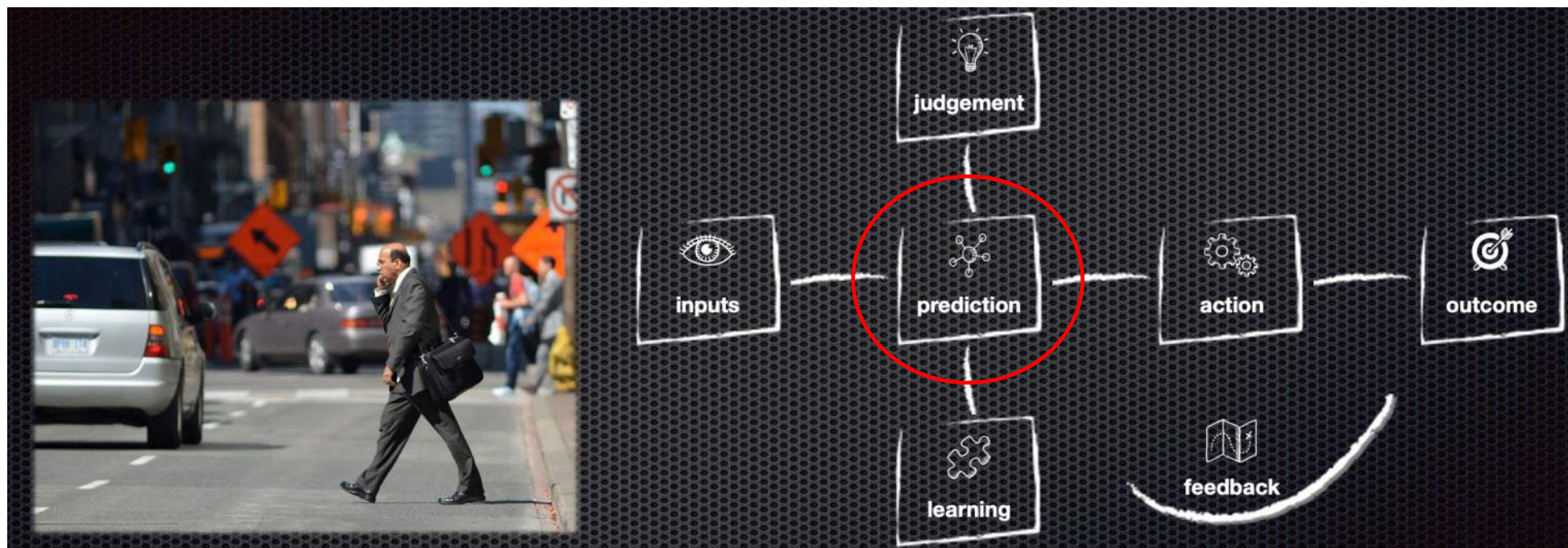


Def: **Physics** is the scientific study of matter, its fundamental constituents, its motion and behavior through space and time, and the related entities of energy and force.

What are we talking about?

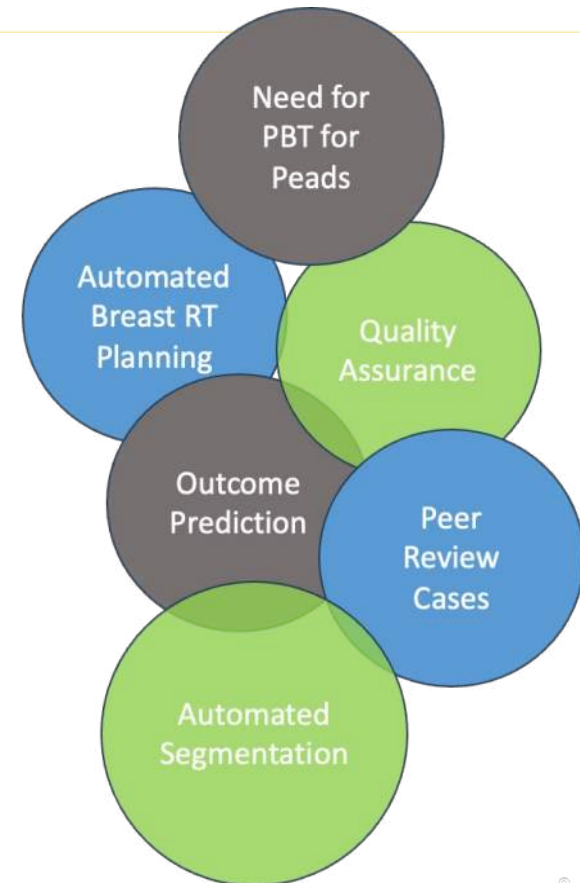


Keep the perspective: AI = Prediction



Why are we (RadOncs) **Al**ming for this?

- We are **nerds obsessed** with Tech
- Limited (RT) **resources**, various (RT) tasks are quite resource intensive
- Operator's **time and expertise impact** quality
- Many processes are **not 'standard'**(izable?)
- The **value of good radiotherapy**
- **Moore's Law**
- **Global impact**, can we (practically) share expertise?
- AI is a **sexy topic** = Automatic Investment



For the Clinic $\neq$ In the Clinic: when theory hits reality



Tension between Accuracy and Interpretability



Machine Prediction meets Human Judgement



Workflow Integration

But humans are very 'rational' (and 'romantic')... until hits home :-P



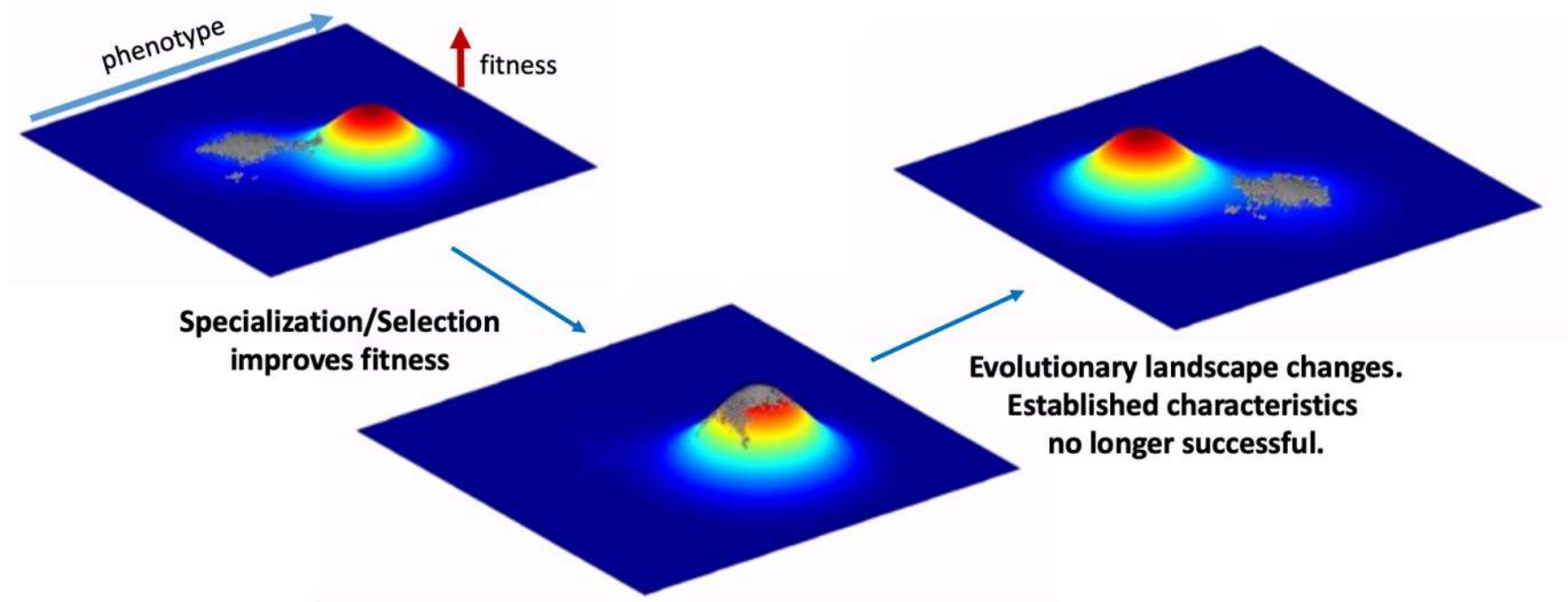
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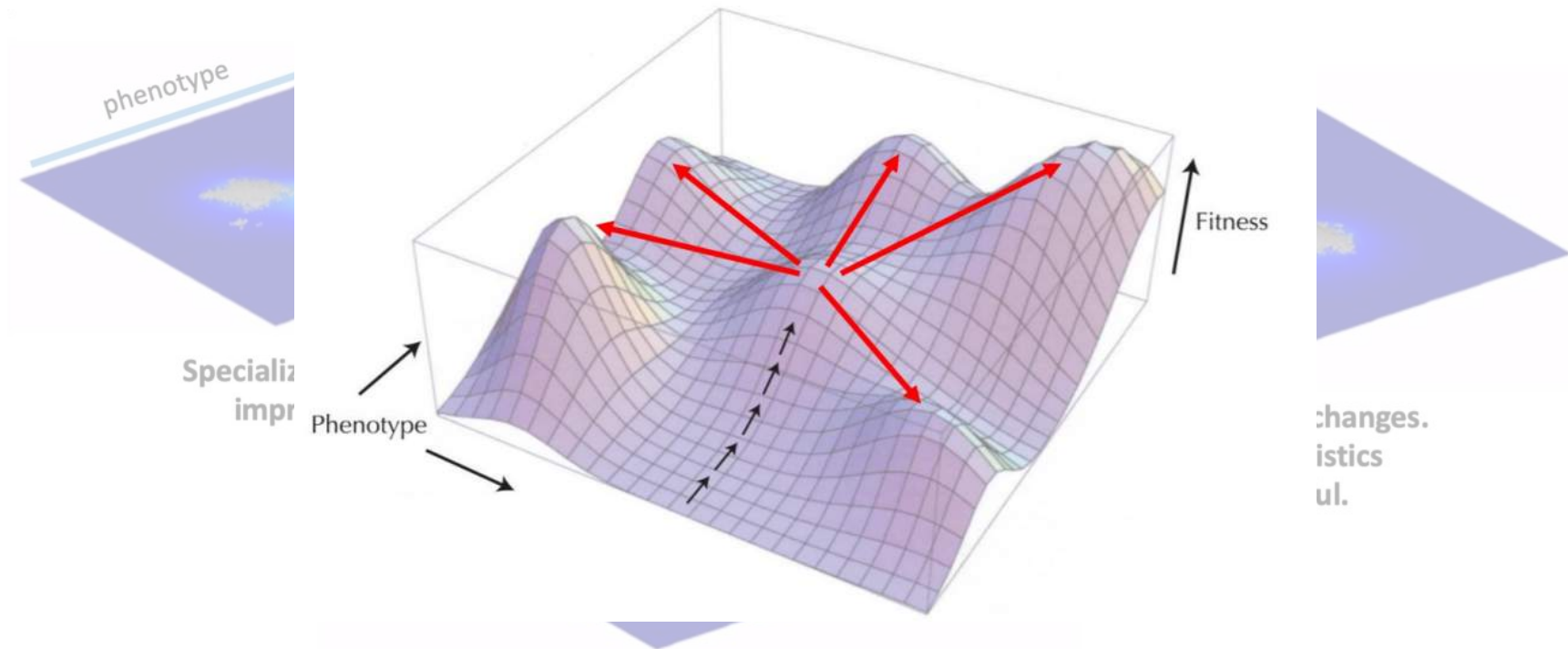
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Diversity reduces extinction risk(s)...



Diversity reduces extinction risk(s)... EXPLORE!

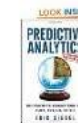


AI (prediction) is Only Getting Better: will change how we operate

(12) United States Patent Spiegel et al.		(10) Patent No.: US 8,271,398 B2
		(45) Date of Patent: Sep. 18, 2012
(54) METHOD AND SYSTEM FOR ANTICIPATORY PACKAGE SHIPPING		
(75) Inventors: Joel R. Spiegel , Woodinville, WA (US); Michael T. McKenna , Bellevue, WA (US); Girish S. Lakshman , Issaquah, WA (US); Paul G. Nordstrom , Seattle, WA (US)		7,006,989 B2 2/2006 Bezos et al. 7,130,803 B1 * 10/2006 Couch et al. 705/1.1 7,222,081 B1 5/2007 Sone 7,610,224 B2 10/2009 Spiegel 7,664,653 B2 2/2010 Dearing 2001/0037316 A1 11/2001 Shiloh 2002/0013744 A1 1/2002 Tsunenari et al. 2002/0138356 A1 9/2002 Dutta et al. 2002/0165729 A1 11/2002 Kuebert et al. 2003/0146138 A1 8/2003 Snapp et al. 2003/0189090 A1 10/2003 Holden 2004/0148217 A1 7/2004 Luring et al. 2004/0153370 A1 8/2004 Yang 2004/0158396 A1 8/2004 Im
(73) Assignee: Amazon Technologies, Inc. , Reno, NV (US)		
(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.		(Continued)
		FOREIGN PATENT DOCUMENTS
(21) Appl. No.: 13/305,611	JP 2002109263 4/2002	
		(Continued)
(22) Filed: Nov. 28, 2011	OTHER PUBLICATIONS	



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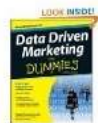
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At the End, Will Become an Ethical Dilemma for Gatekeepers



So... you're suggesting that we should deprive patients of a [insert here] just because the source of it is different from what we're accustomed to*?

** granted all other evaluations, QA, checks and balances, etc. remain in place*

True North: what do I want as a clinician (or patient)?

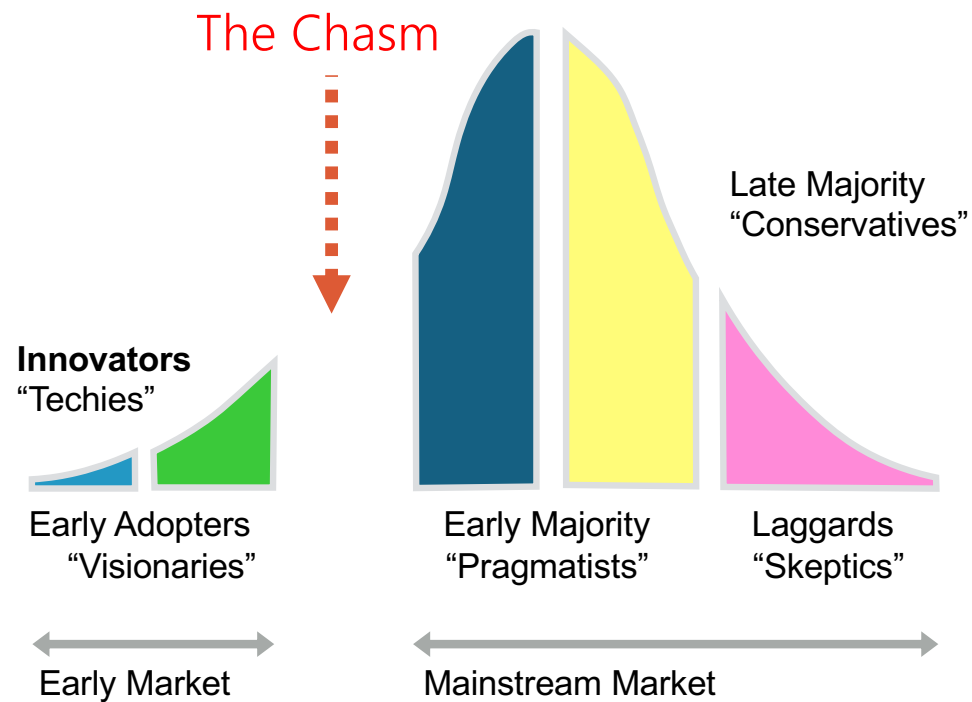
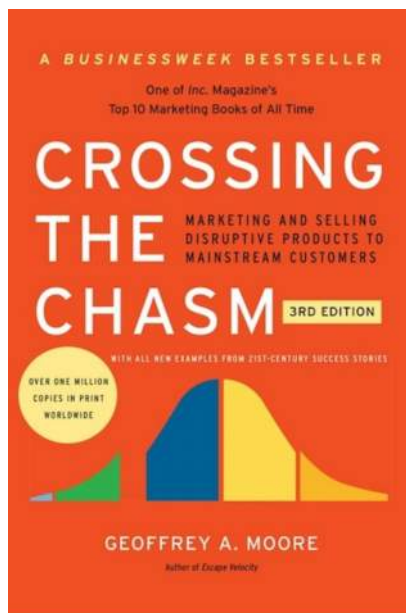
- **Fundamental goal** is to improve value for patients. Delivering high value is the definition of **success**.

$$\text{VALUE} = \frac{\text{Set of outcomes that matter for a specific condition}}{\text{Total cost of delivering them over the full care cycle}}$$

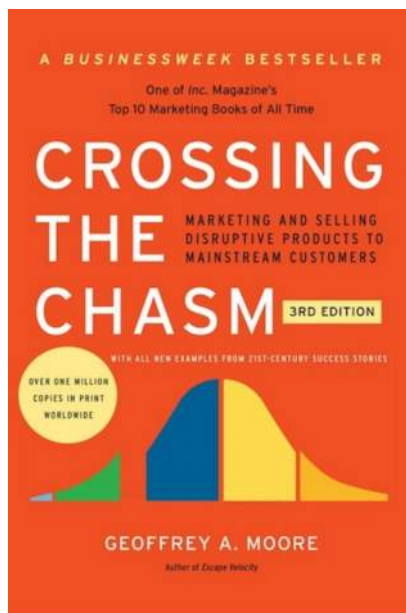
- Culture will lead towards **multidisciplinary care**, aligning and re-organizing into **truly integrated practice** units.
- Validate **value-enhancing** strategies, including resource use **optimization**.
- Essential components of forthcoming value-based **reimbursement** models.
- The most powerful single lever for reducing cost is **improving outcomes**.

Automated Planning Clinical Deployment and Evaluation

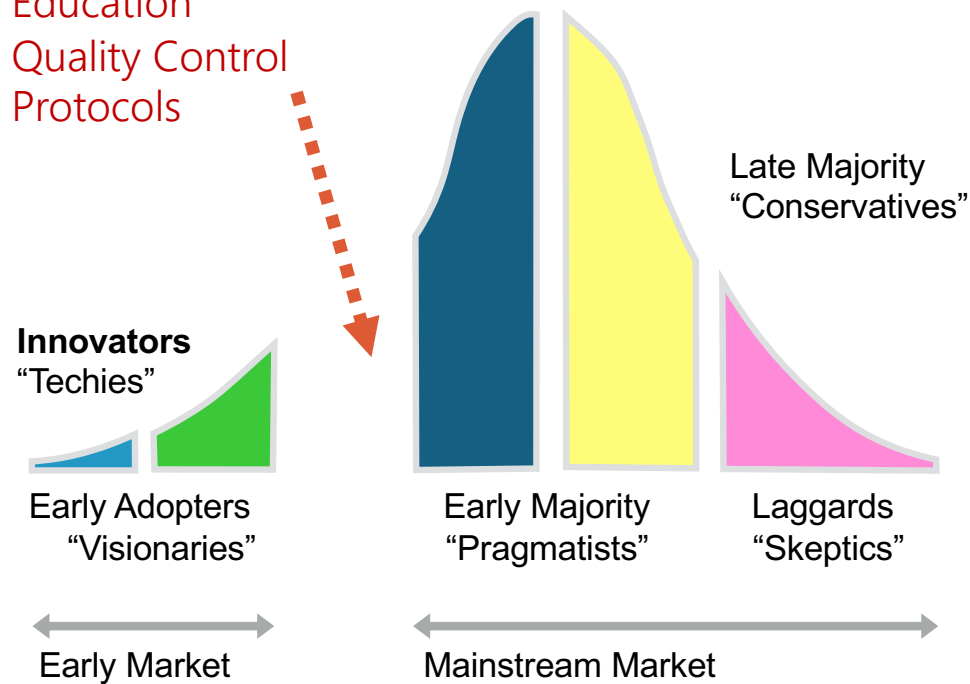
Technology Adoption Life Cycle



AI Adoption Life Cycle



- Trust
- Education
- Quality Control
- Protocols





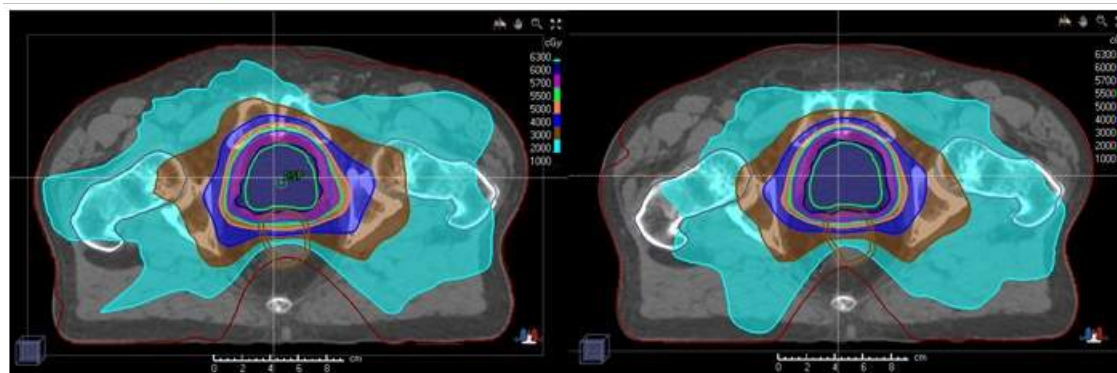
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LETTERS

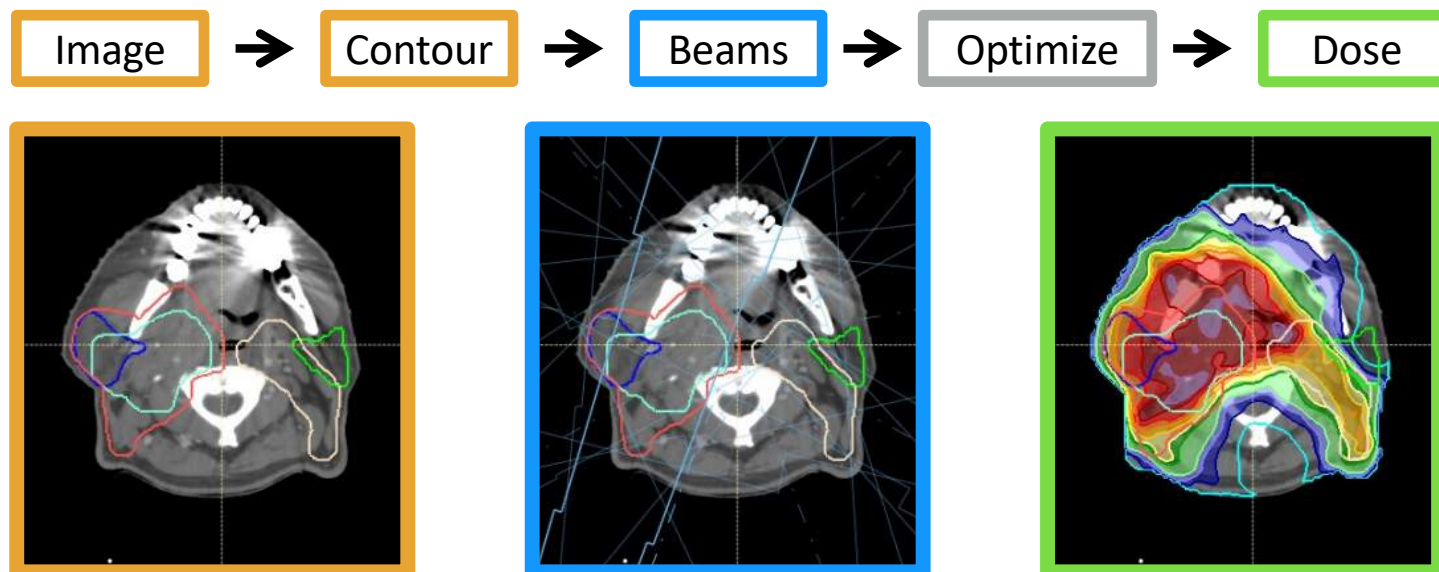
<https://doi.org/10.1038/s41591-021-01359-w>

Clinical integration of machine learning for curative-intent radiation treatment of patients with prostate cancer

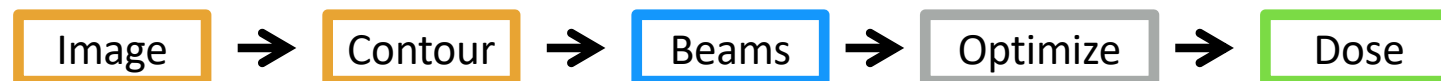
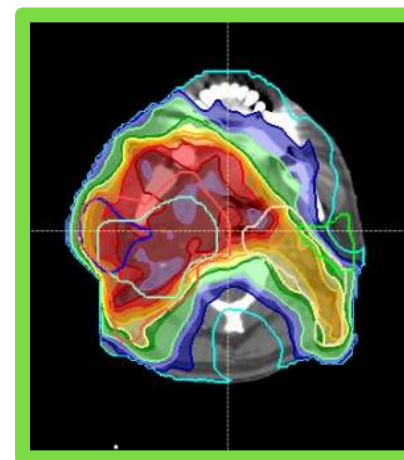
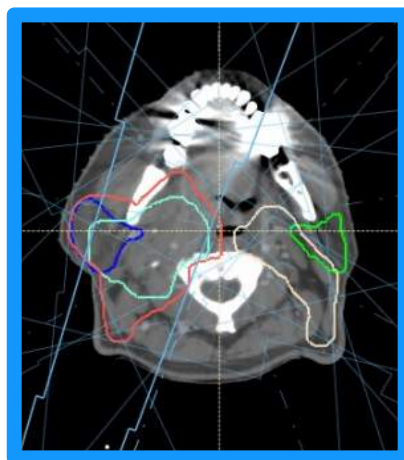
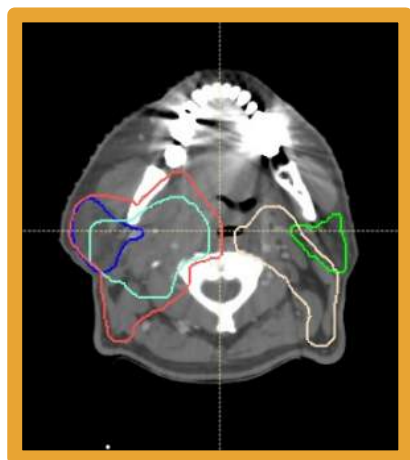
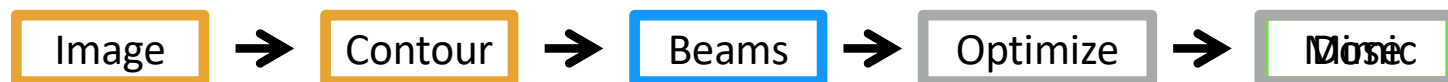
Chris McIntosh ^{1,2,3,4,5,6,8}, Leigh Conroy^{1,2,7,8}, Michael C. Tjong^{1,7}, Tim Craig^{1,2,7}, Andrew Bayley^{1,7}, Charles Catton^{1,7}, Mary Gospodarowicz^{1,7}, Joelle Helou^{1,7}, Naghmeh Isfahanian ^{1,7}, Vickie Kong^{1,7}, Tony Lam^{1,7}, Srinivas Raman^{1,7}, Padraig Warde^{1,7}, Peter Chung^{1,7}, Alejandro Berlin ^{1,2,7,8}  and Thomas G. Purdie ^{1,2,6,7,8} 



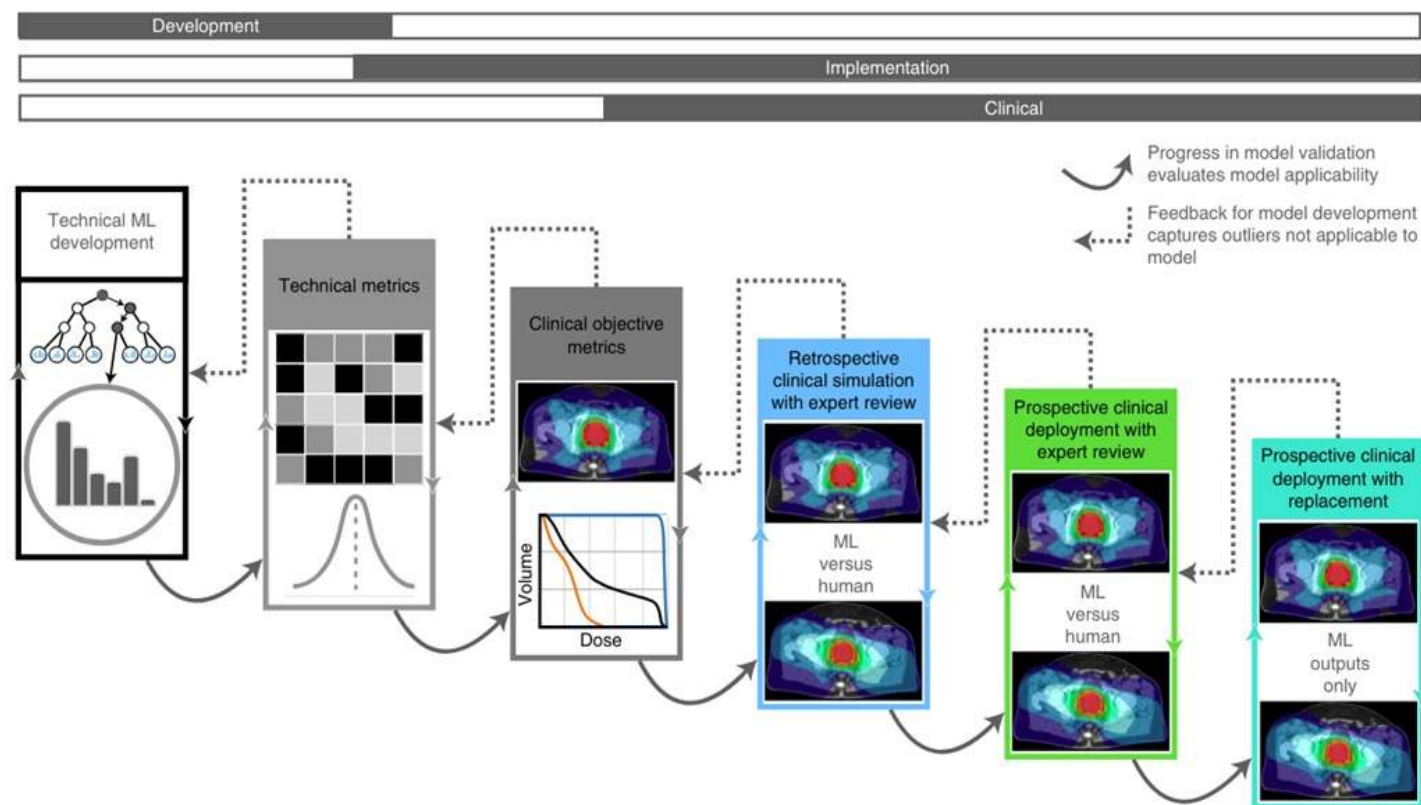
The Method



The Method

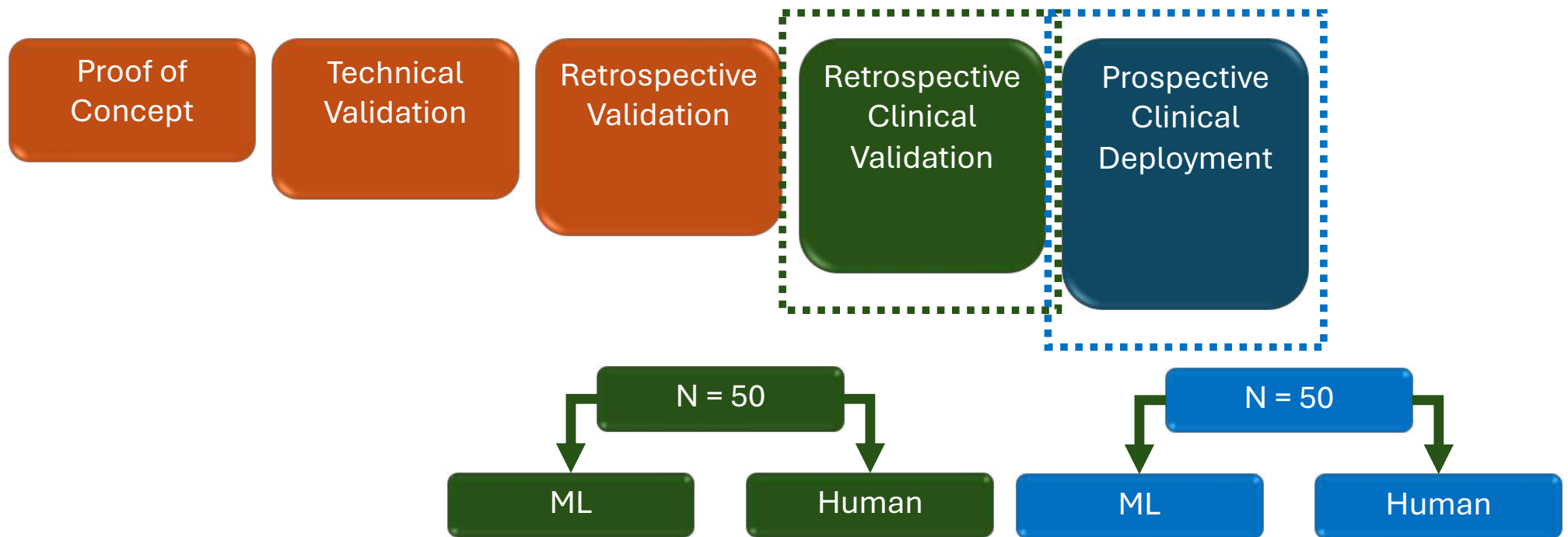


Framework



McIntosh, Conroy, et al., Nat Med 2021 Jun;27(6):999-1005.

ML Clinical Validation Framework



Plan Comparison. Which plan has better: *

	Prostate_A1	Prostate_A2	They are Equivalent
Target Coverage	☐	☐	☐
OAR Sparing	☐	☐	☐
High Dose Conformity	☐	☐	☐
Rectal Dose Gradient	☐	☐	☐
Lateral Dose Symmetry	☐	☐	☐

Plan Acceptability *

	Acceptable	Unacceptable
Prostate_A1	☐	☐
Prostate_A2	☐	☐

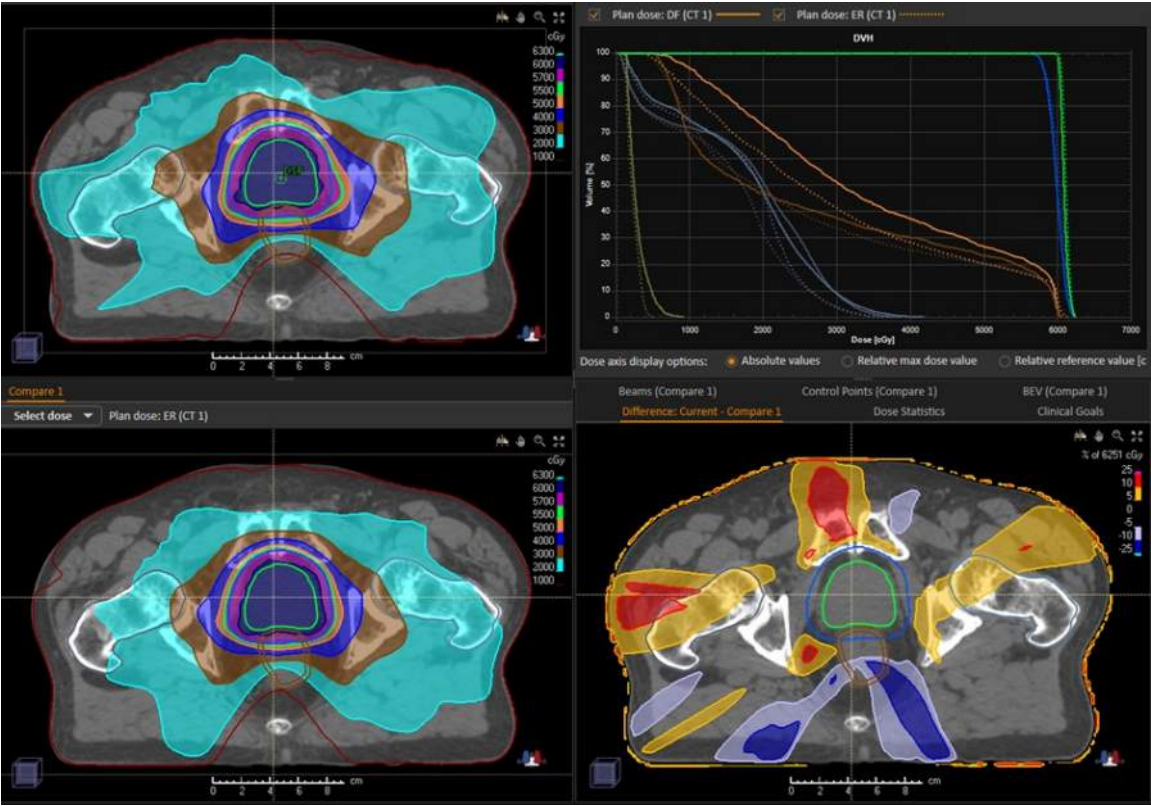
Preferred Plan *

	Preferred Plan
Prostate_A1	☐
Prostate_A2	☐
Equivalent → go with Prostate_A1	☐
Equivalent → go with Prostate_A2	☐
Do not like either plan	☐

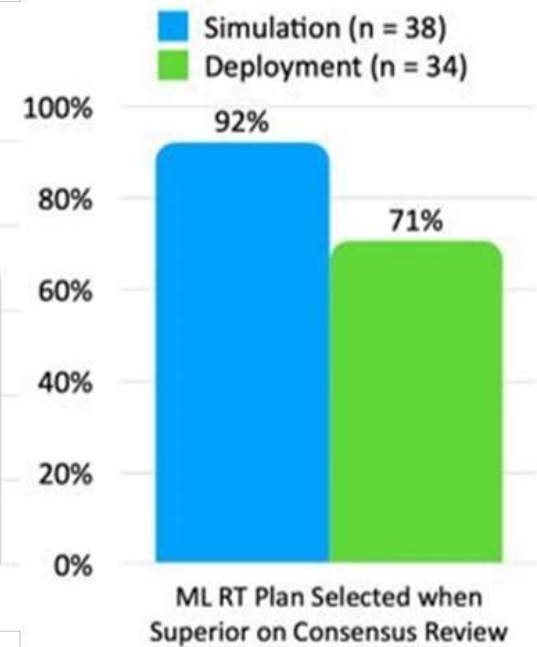
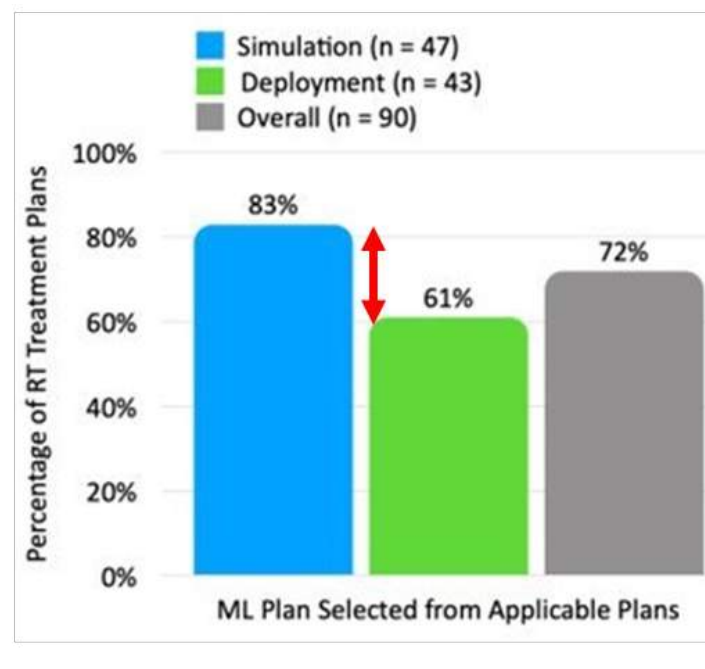
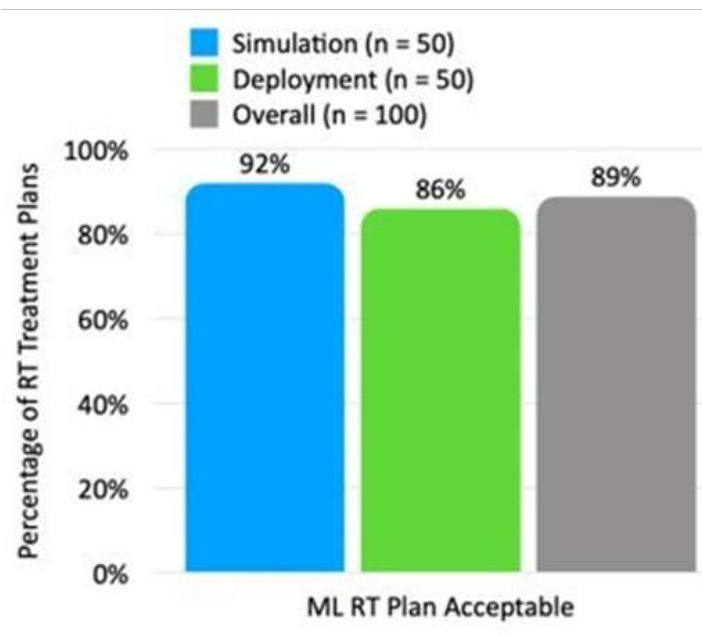
Which plan do you think is the Automated Plan? *

	Automated Plan
Prostate_A1	☐
Prostate_A2	☐

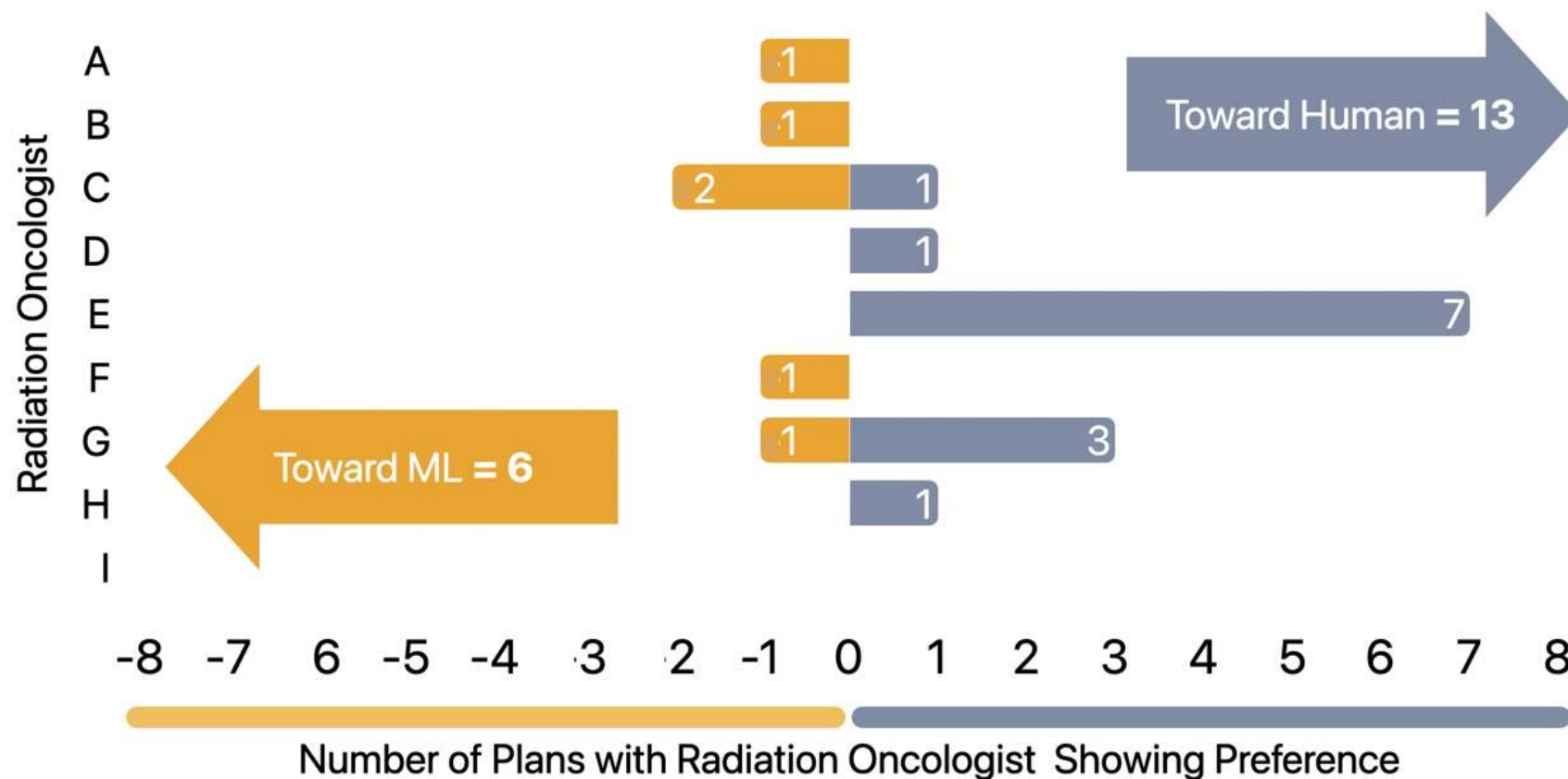
Submit



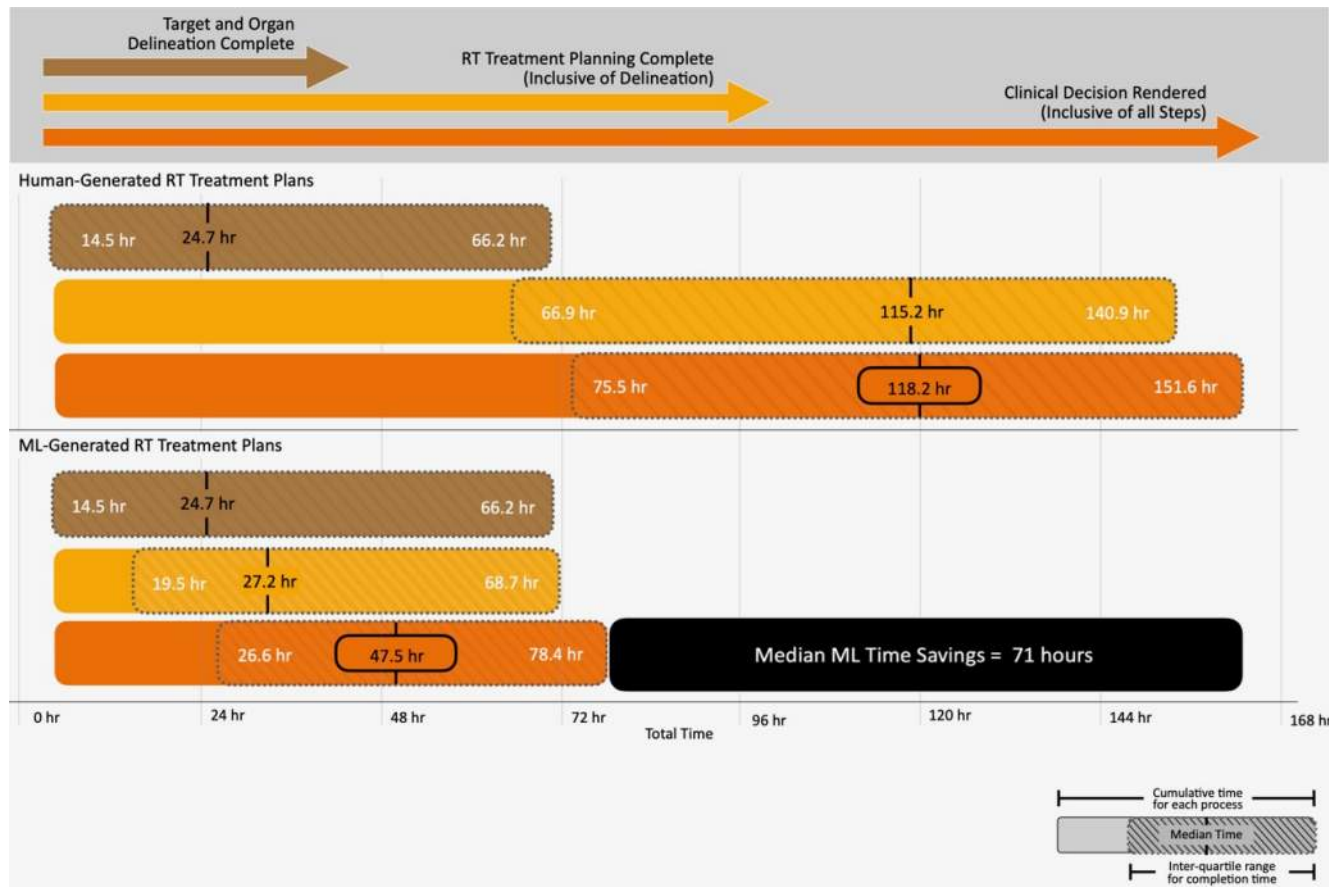
Simulation versus Real-World Acceptance



Radiation Oncologist Preference Toward ML or Human?



Efficiency Simulation

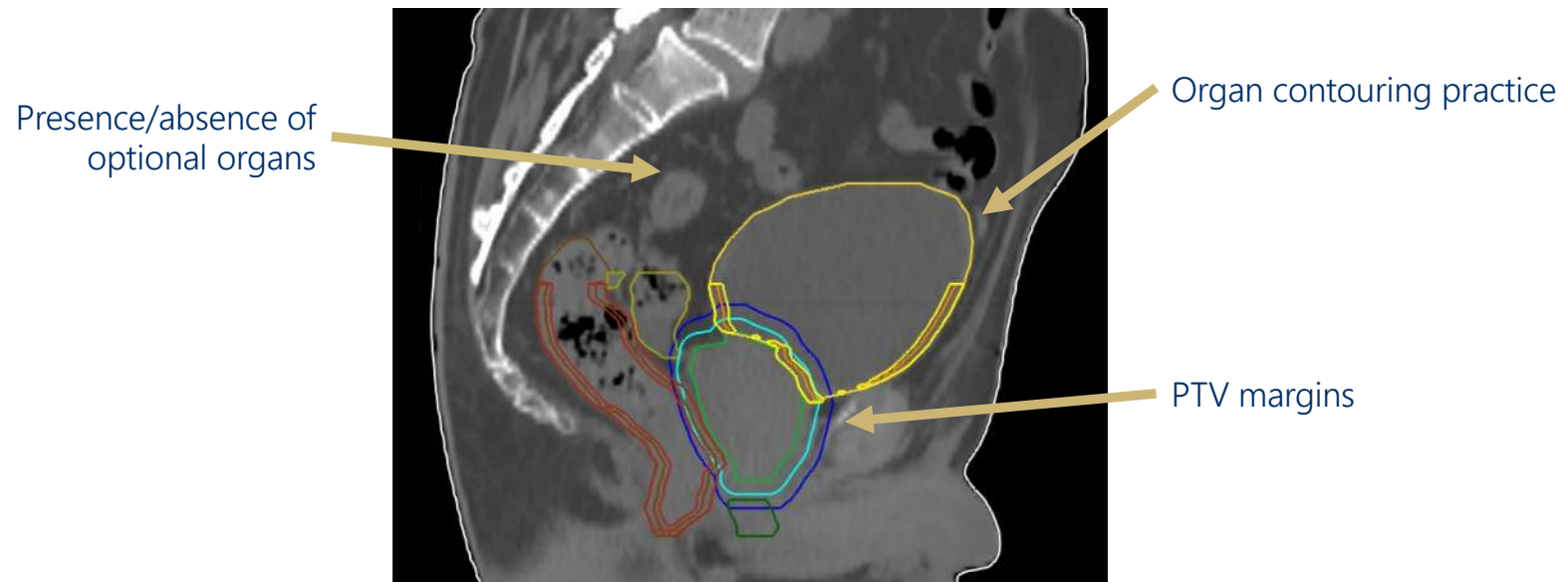


Automation = Loss of Domain Knowledge?

	Current Role	AI Role
Dosimetrist	• Create patient plans	• "Repetitive tasks will be taken over by automation and enhanced roles will emerge focusing on tasks that require human judgement and QA" • Curate and create plans for AI training • Evaluating and adjusting AI plans
Medical physicist	• TPS commissioning & QA • Patient-specific chart check	• Decision making & maintenance of AI • QA implementation, maintenance, and oversight of AI."
Radiation Oncologist	• Prescribing, contouring	• Evaluating AI contours and plans • Treatment decisions (including need for non-AI alternate plan)

Model Stability with Change in Input

- Test impact of input data variations on algorithm performance
- Benchmarked against ground truth “Base” plan
- Model of routine quality assurance



Conroy et al., Phys. Med. Biol. 66 (2021) 134001

Clinical Practice Impacts Performance

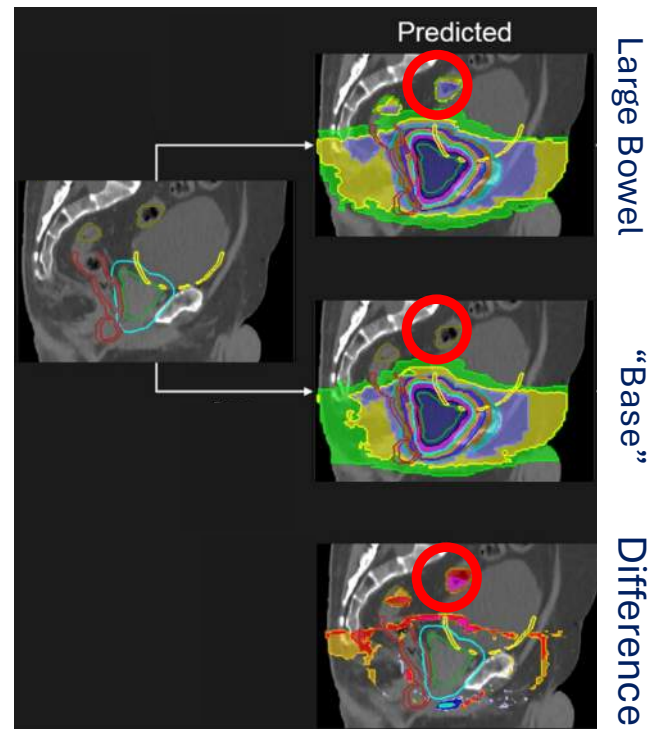
Bowel contours are usually only included in clinical plans when they are close to the target



All plans with bowel contours in the training set have dose to bowel



Model places dose in bowel contour, even when distant!



Conroy et al., Phys. Med. Biol. 66 (2021) 134001

Clinical Evolution and Outcomes

Changing Clinical Practice @ Princess Margaret

- Bladder and rectum wall changes – 2D to 3D walls
- PTV margin reduction – 10 mm (7 mm post) → 7 mm uniform → 6 mm uniform
- 60 Gy in 20 used for more complex treatments
- Dosimetrist empowered to:
 - Tweak the plan with an additional optimization
 - Create a human plan with conventional approach
- Challenging to establish ML performance in realistic clinical scenario

ML Clinical Validation Framework

Proof of
Concept

Technical
Validation

Retrospective
Validation

Retrospective
Clinical
Validation

Prospective
Clinical
Deployment

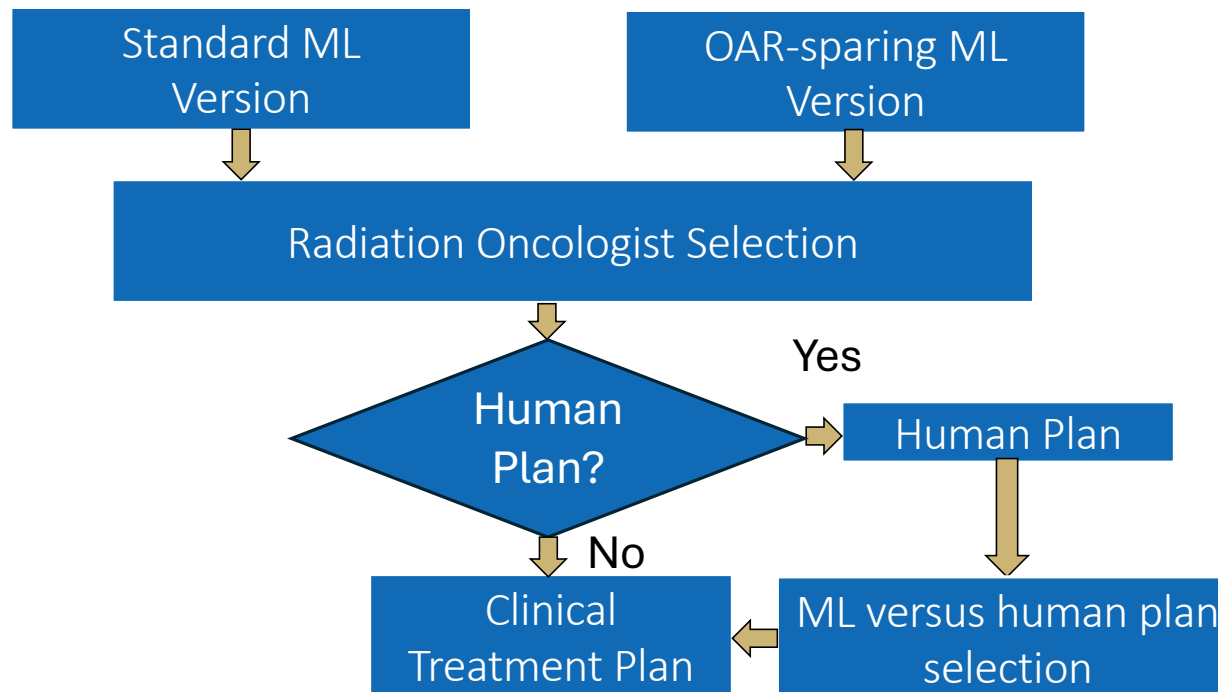
Human-in-
the-loop
&
Clinical
Outcomes
Assessment

Clinical Outcomes Assessment

- Patients treated with 7 mm PTV margin using ML model trained with 10 mm margin (7 mm post)
- Collected physician-recorded GU & GI toxicities
- Toxicity-free survival Kaplan-Meier analysis based on log-rank tests
- Compared Grade 2+ GI & GU toxicities for ML vs. human-generated plans
- Follow up = 34 (range 6 – 51) months

Under Review

Methods – Study Design

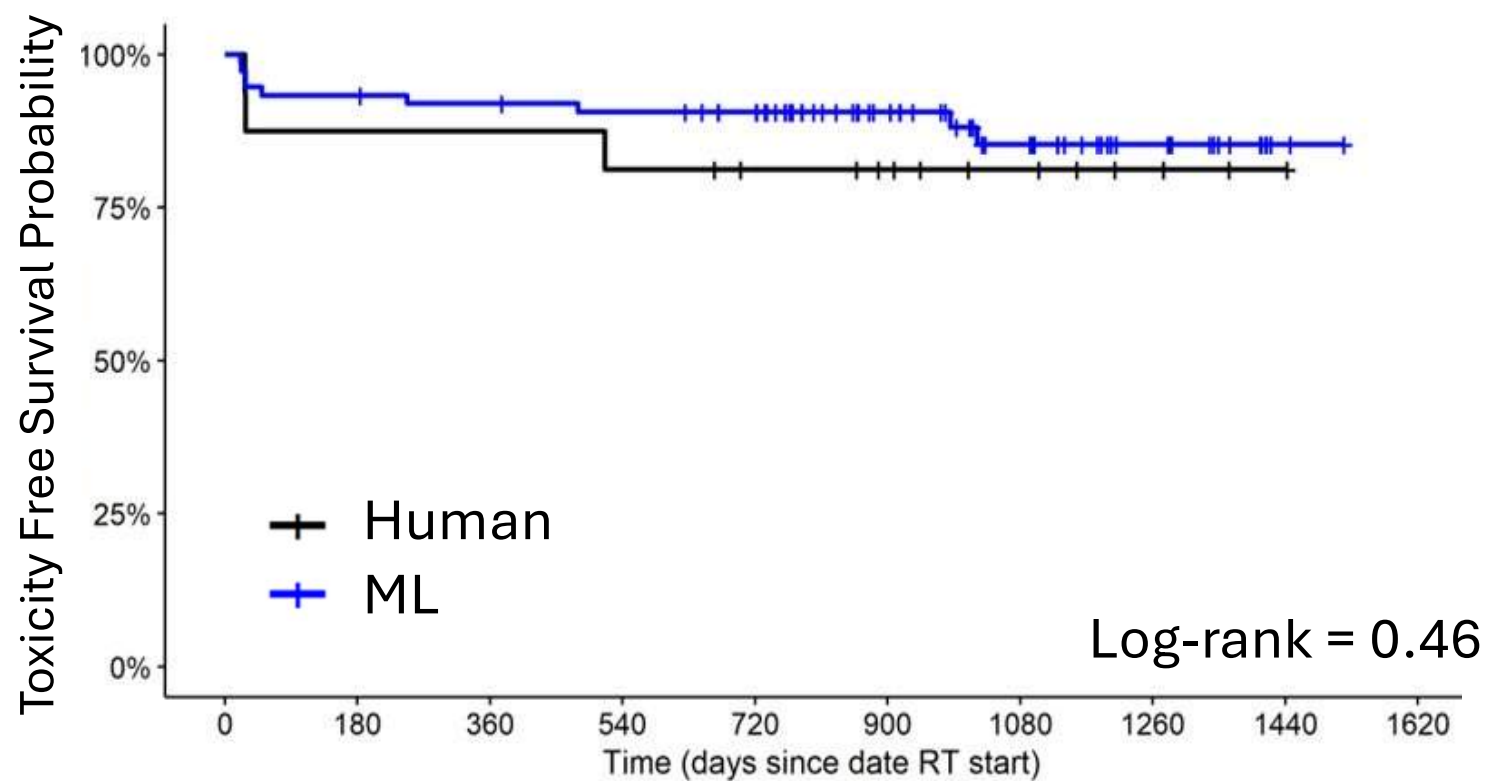


Results

- Human-in-the-loop radiation oncologist plan selection for treatment:
 - **ML plan = 82 % (76/93) cases**
 - **Human plan = 18 % (17/93 cases)**
- No GI G2+ toxicities

Under Review

Results – Grade 2+ GU Toxicities



Under Review

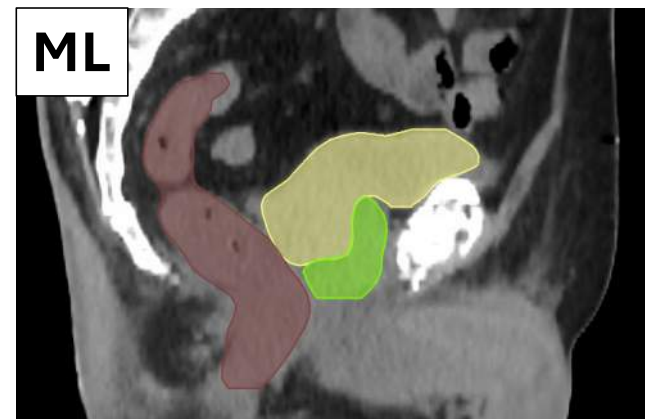
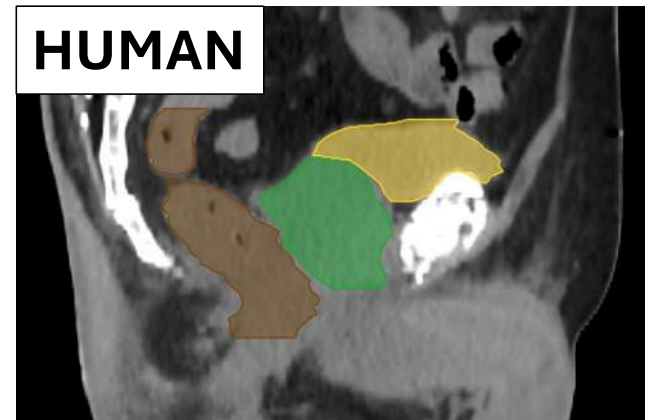
Discussion/Conclusions

- Human-in-the-loop ML planning **82 %** ML plan selection exceeded rate of **61 %** selected in prospective phase of previous study
- No G2+ GI toxicity observed for either group and **ML** versus **human** plan toxicity-free survival for **G2+ GU not significantly** different
- ML planning maintains low toxicity in curative-intent prostate cancer RT and not statistically different from human planning

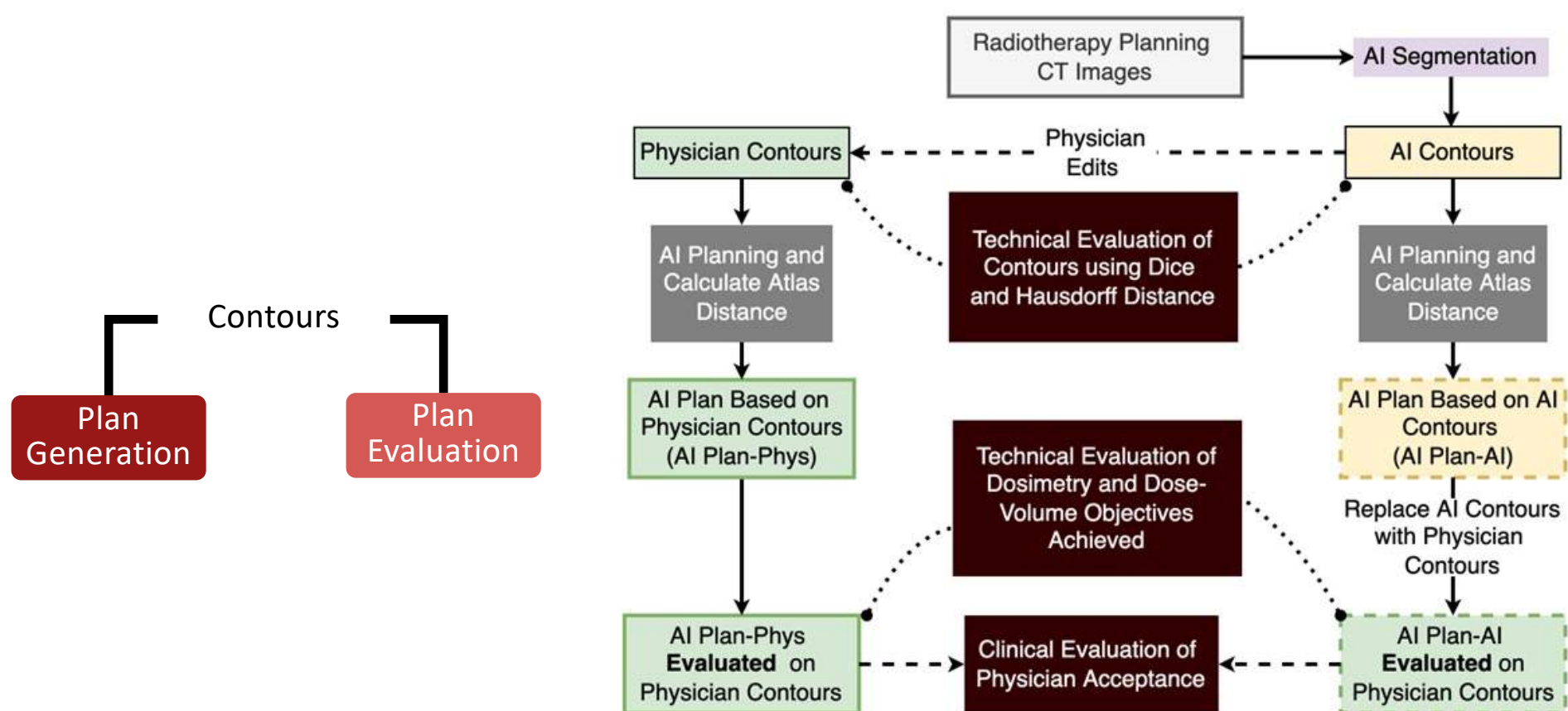
Under Review

Male Pelvis ML Contour Implementation

- Benchmarking:
 - Curated and edited manual contours
 - Measured ML contouring time
 - Geometric analysis – DICE & DTA
 - Qualitative assessment
 - 20 prostate cases, 10 prostate bed
- Implementation:
 - Save original ML contours for all patients
 - Planning script requires CTV edit
 - Compared ML versus clinical contours in peer-review rounds



Evaluation of E2E ML planning pipeline

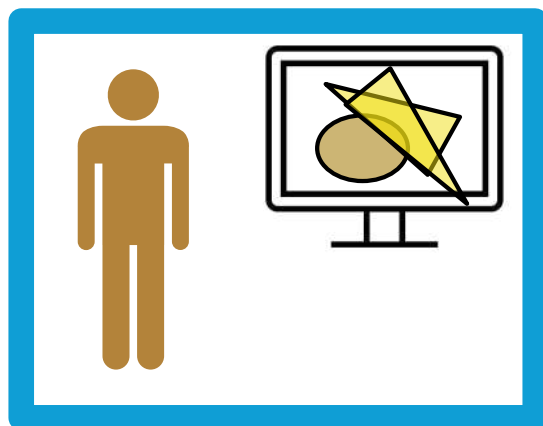


ML for Changing Clinical Practice Deployment Strategies

ML for Clinical Practice Change in Partial Breast

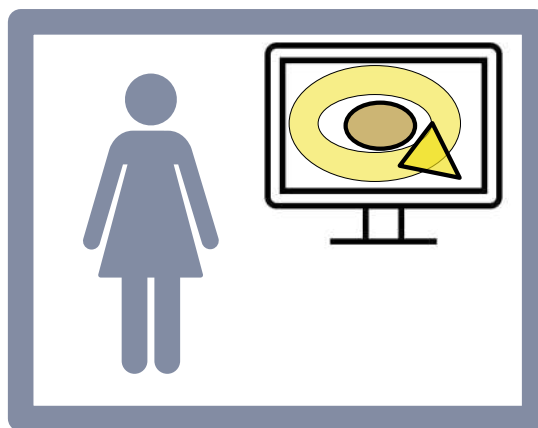
Current

Manual Fixed Beams



Future ?

ML Arc Beams



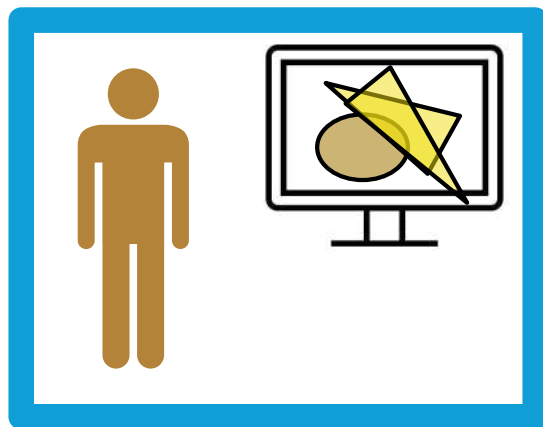
Not
changing:

- ⇒ Dose Fractionation
- ⇒ Expansion Margins
- ⇒ Localization Imaging

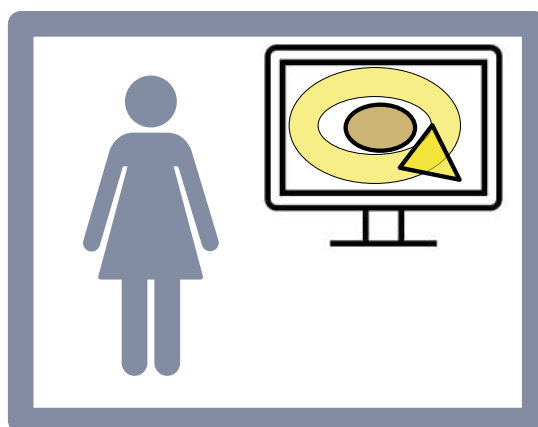
ML for Clinical Practice Change in Partial Breast

Current

Manual Fixed Beams



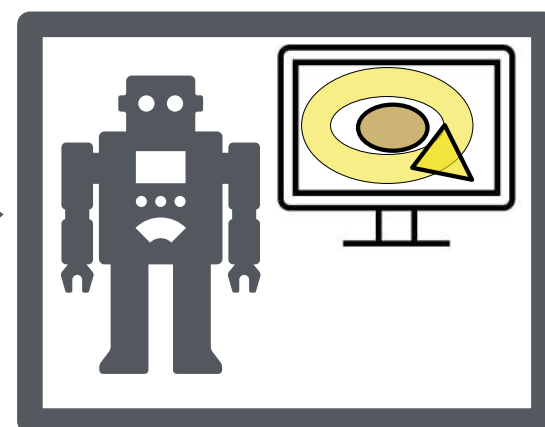
Manual Arc Beams



Conventional

Final

ML Arc Beams



Collect
Training Data



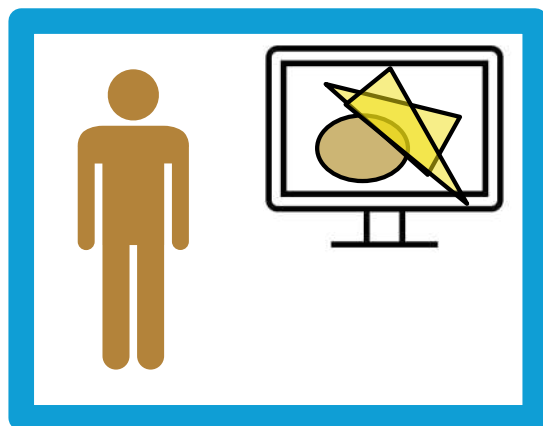
Train ML

ML for Clinical Practice Change in Partial Breast

Practice Change Approach $\Rightarrow$
Training Data Curation Creation

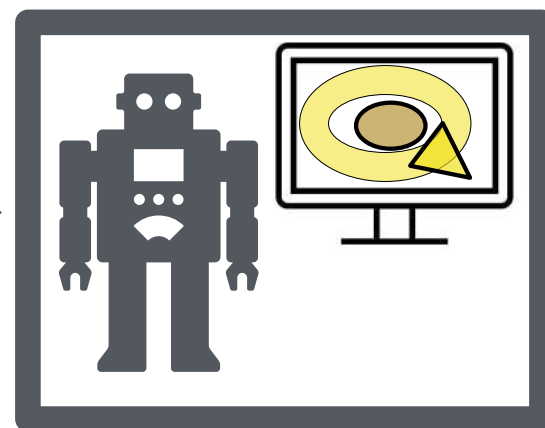
Current

Manual Fixed Beams



Final

ML Arc Beams



Expert Data
Curation

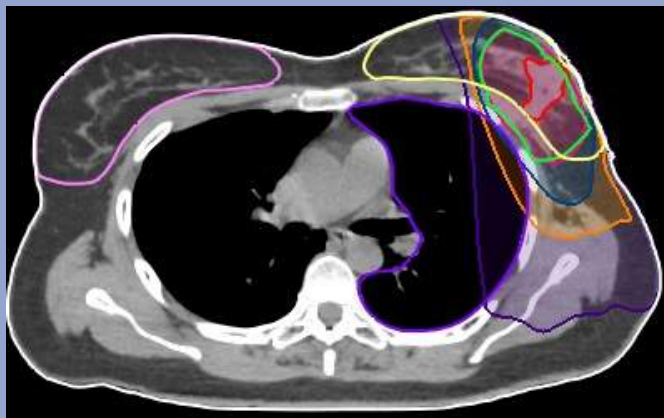


Train ML

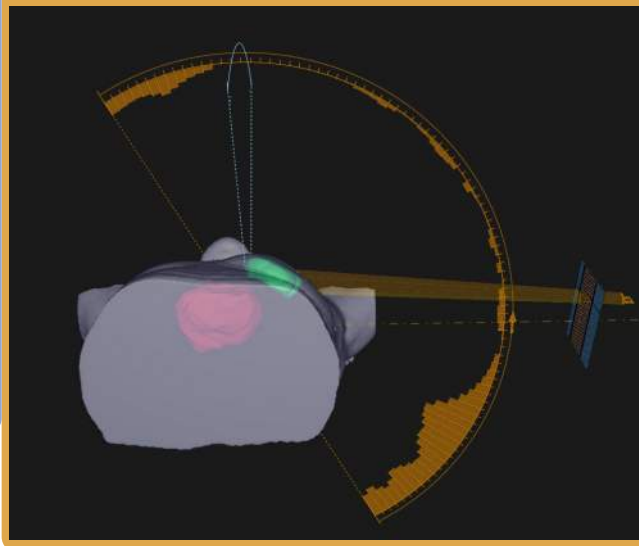
ML for Clinical Practice Change in Partial Breast

Training Data Creation

Contouring



Arc Geometry

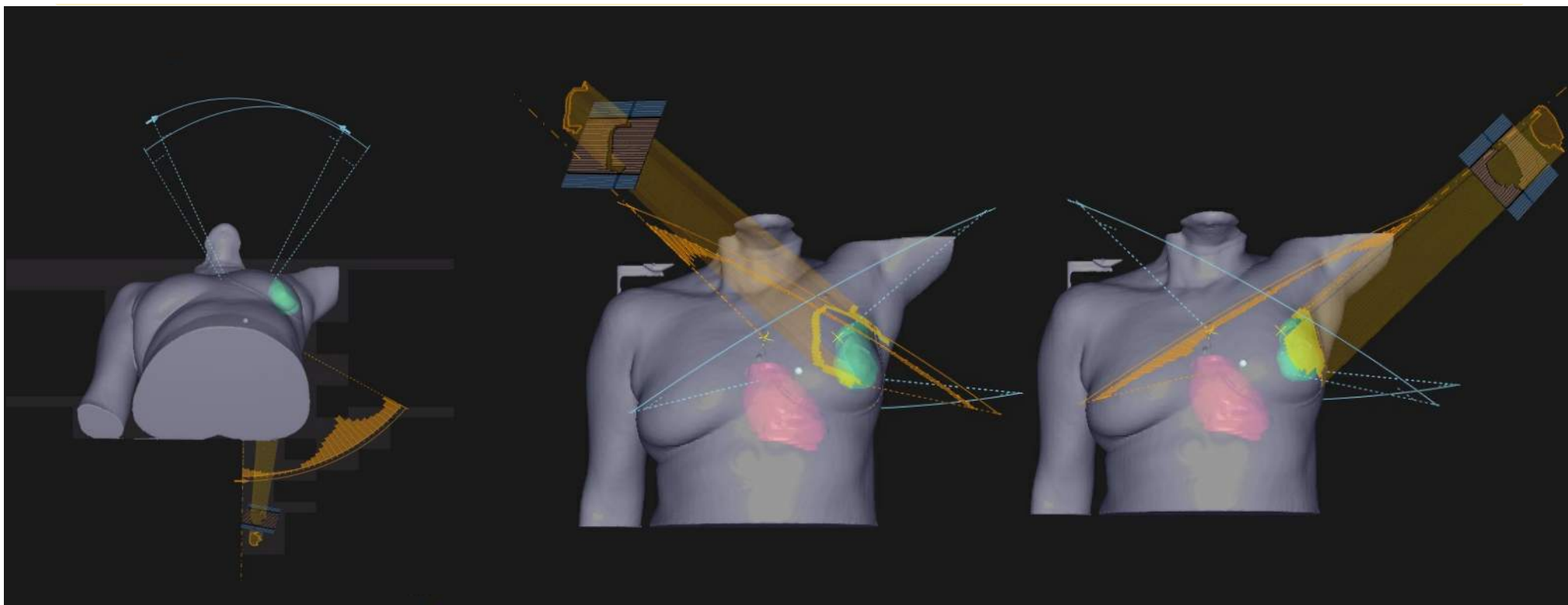


Dosimetric Clinical Goals

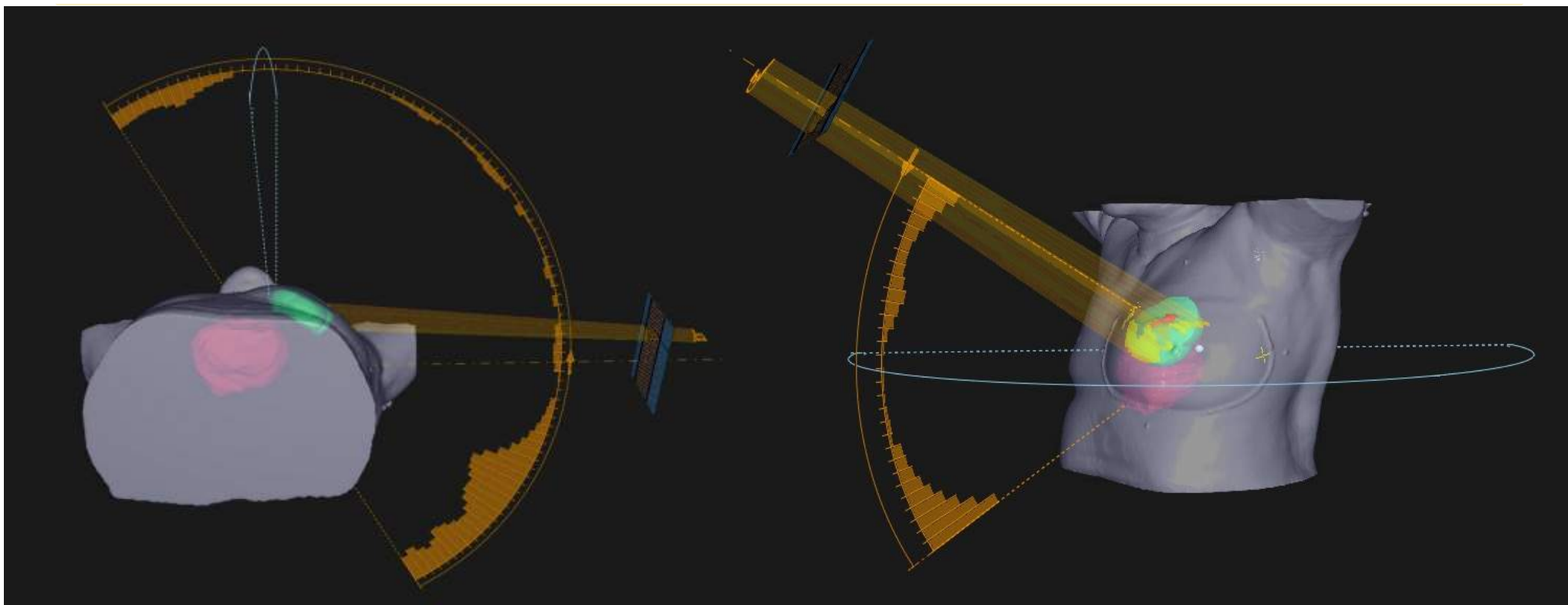
TABLE 1 Planning constraints (minor variations) used in this study compared to those from phase III trials

Planning constraints for this study ³⁰	NSABP B39/RTOG 0413 ³	RAPID ¹²	Florence ⁷
DEV and prescription			
D98% ≥ 95%	V90% ≥ 90%	V95% = 100%	V95% = 100%
27 Gy/5	38.5 Gy/10 BID	38.5 Gy/10 BID	30 Gy/5
PTV			
CI < 1.2 (1.2–1.4)			
D ₁₀₂ < 107%	D _{max} < 120%	D ₁₀₂ < 107%	D _{max} < 105%
Ipsilateral breast			
V50% < 40% (40%–60%)	V50% < 60% (60%–65%)	V50% < 50% (50%–65%)	V50% < 50%
V95% < 15% (15%–25%)	V100% < 35% (35%–40%)	V95% < 25% (25%–35%)	
Contralateral breast			
V3% < 3%	D _{max} < 3%	D _{max} < 3%	D _{max} < 3%
Ipsilateral lung			
V10% < 20% (20%–25%)		V10% < 20% (20%–25%)	
V30% < 10% (10%–13%)	V30% < 15% (15%–20%)	V30% < 10% (10%–13%)	V30% < 20%
Heart (right-sided)			
V5% < 5% (5%–8%)	V5% < 5% (5%–10%)	V5% < 5%	V10% < 10%
Heart (left-sided, lower inner quadrant)			
V15% < 5% (5%–8%)	V5% < 40% (40%–45%)	V15% < 5%	V10% < 10%
Heart (left-sided, other quadrants)			
V10% < 5% (5%–8%)	V5% < 40% (40%–45%)	V10% < 5%	V10% < 10%

ML for Clinical Practice Change in Partial Breast – Lateral Beams



ML for Clinical Practice Change in Partial Breast – Medial Beams



Simulated Training Data

Benefits	Risk	Implication	Mitigation
Highly curated data	Bias	Not all clinical scenarios	⇒ Consecutive clinical patients ⇒ Multidisciplinary expert review
Planned for ML training	Bias	Variation in clinical preferences	Early and iterative feedback from larger group
Skip manual arc beam treatment planning	Domain knowledge	Experience gap when ML fails	Train multidisciplinary group of knowledge experts

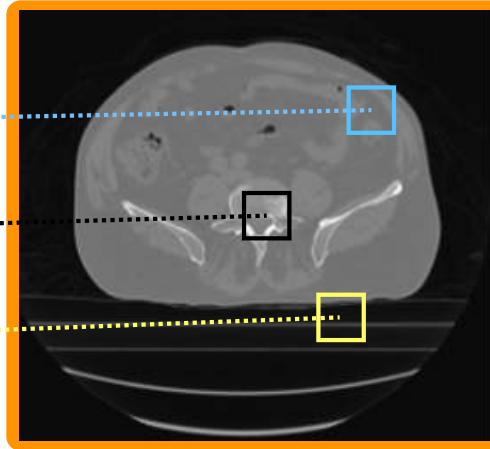
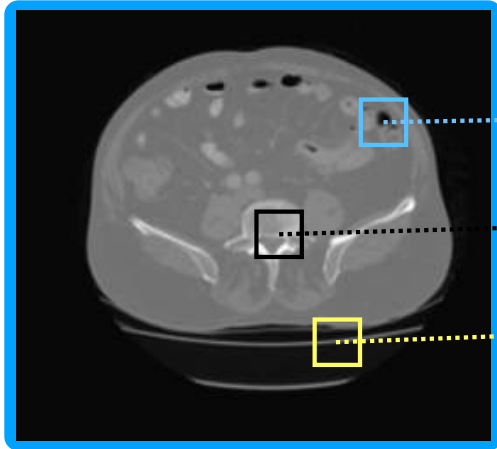
AI to Collapse the Time Between Diagnostic Imaging and Treatment

Generating Synthetic Images from Diagnostic Images

Diagnostic Imaging



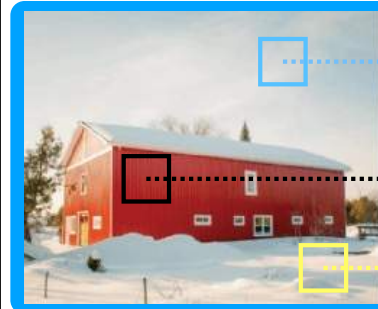
Radiation Therapy Imaging



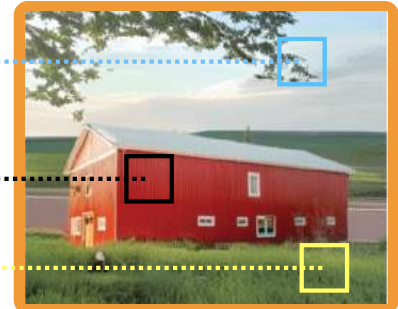
Example:

AI Learns How to Make Winter Look Like Summer

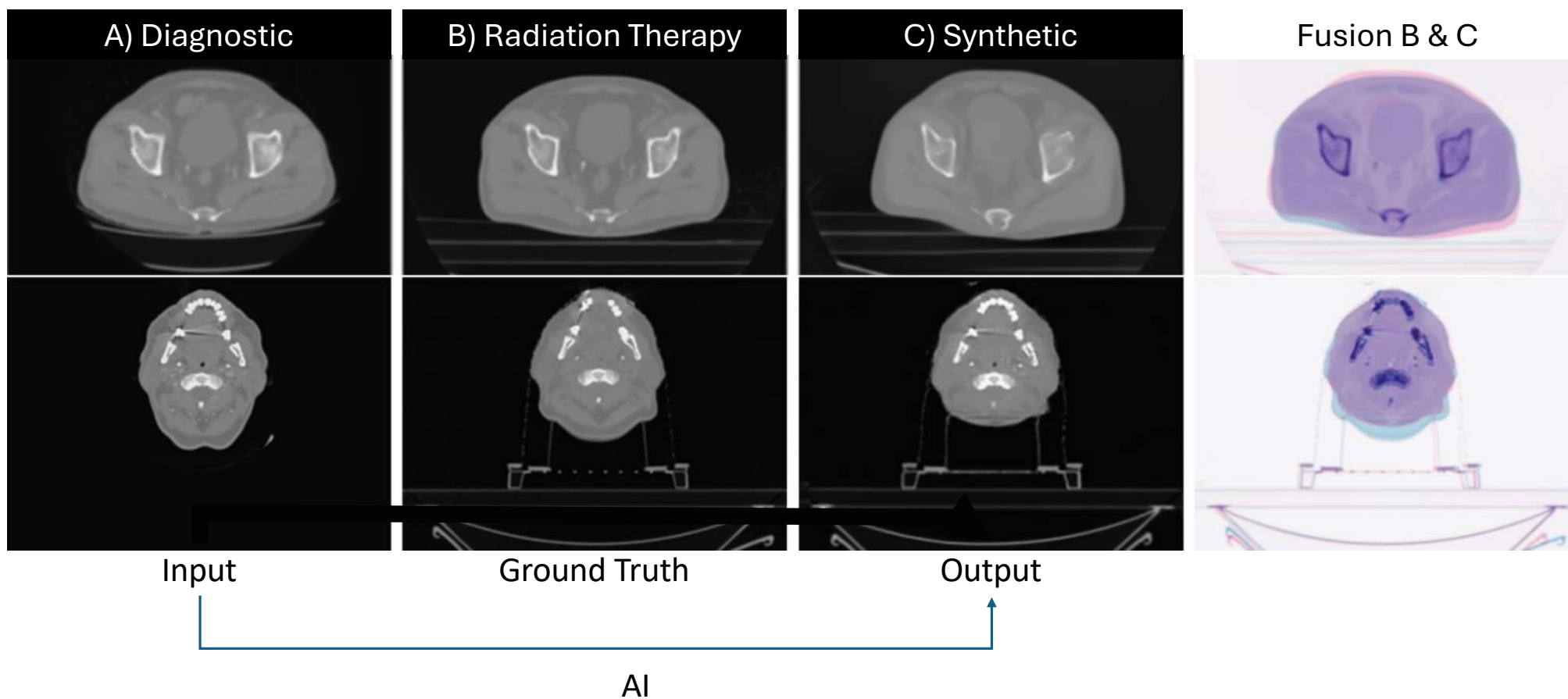
Winter



Summer



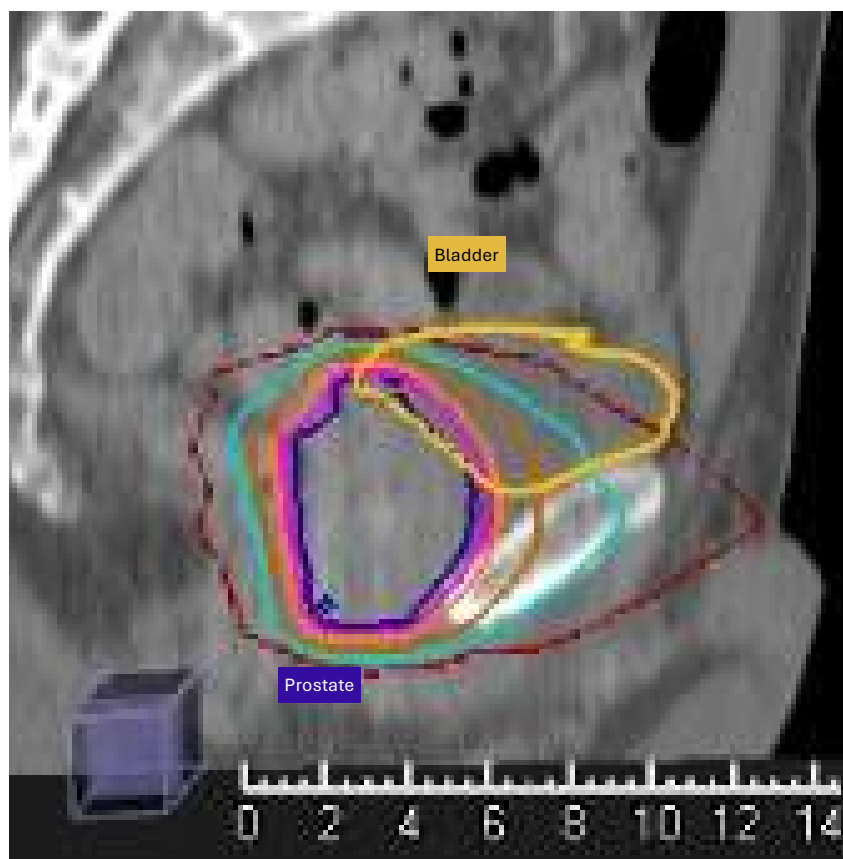
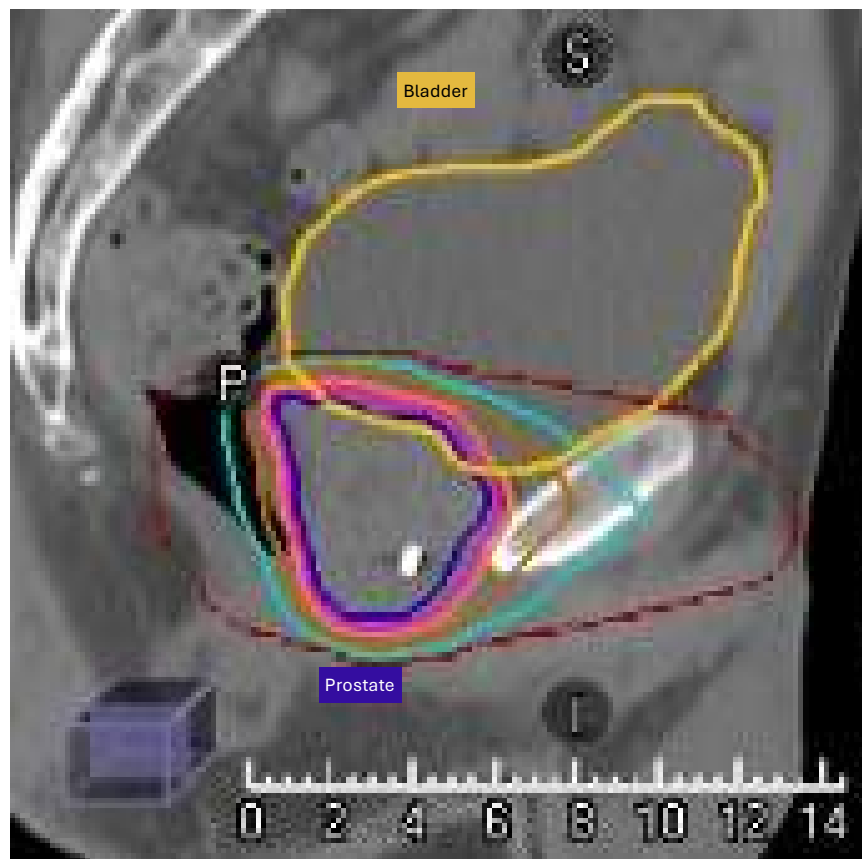
Generating Synthetic Images from Diagnostic Images



Generating Synthetic Images from Diagnostic Images

Diagnostic

Radiation Therapy



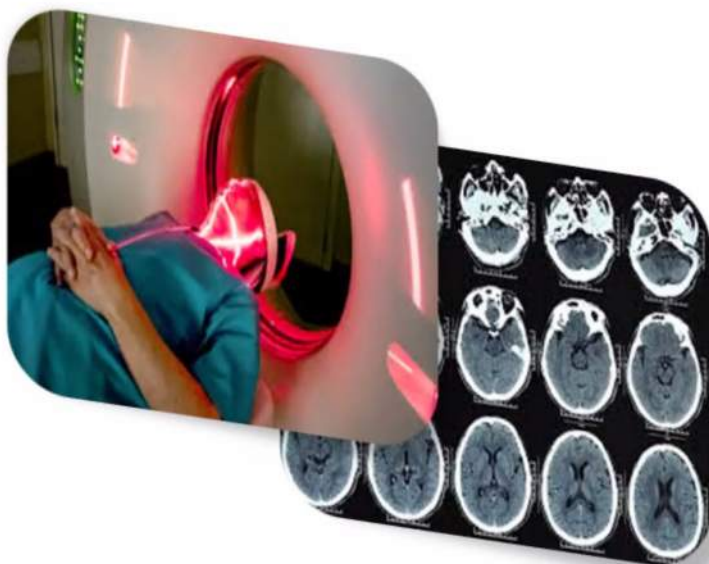
Generating Synthetic Images from Diagnostic Images

Prostate anatomy	<u>Clinical Objective</u> dose @ volume% (cGy)	mean dose $\pm$ sd (cGy)		
		Diagnostic	Radiation Therapy	Synthetic
Target	dose @ 99% > 6000	5717 $\pm$ 13	5725 $\pm$ 7	5716 $\pm$ 15
Rectum	dose @ 30% < 4700	2877 $\pm$ 615	2761 $\pm$ 732	2728 $\pm$ 610
Bladder	dose @ 30% < 4700	3411 $\pm$ 652	3401 $\pm$ 653	<u>2688 $\pm$ 370</u>

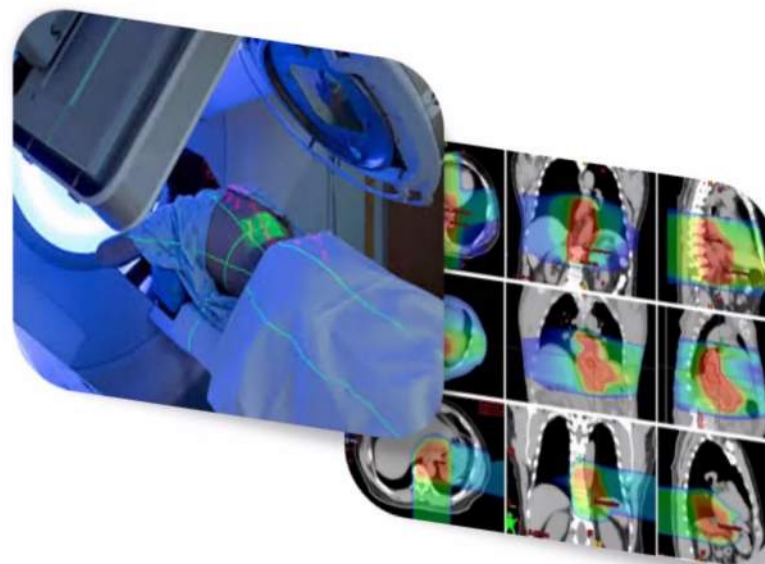
(Directly) Challenging the Workflow

Compartmentalization: ionizing radiation

Diagnostic purposes



Therapeutic purposes



The (problem of) waste!

Current Workflow



The Proposal: focus inward or outward?!

Current Workflow



Symptoms
& Signs

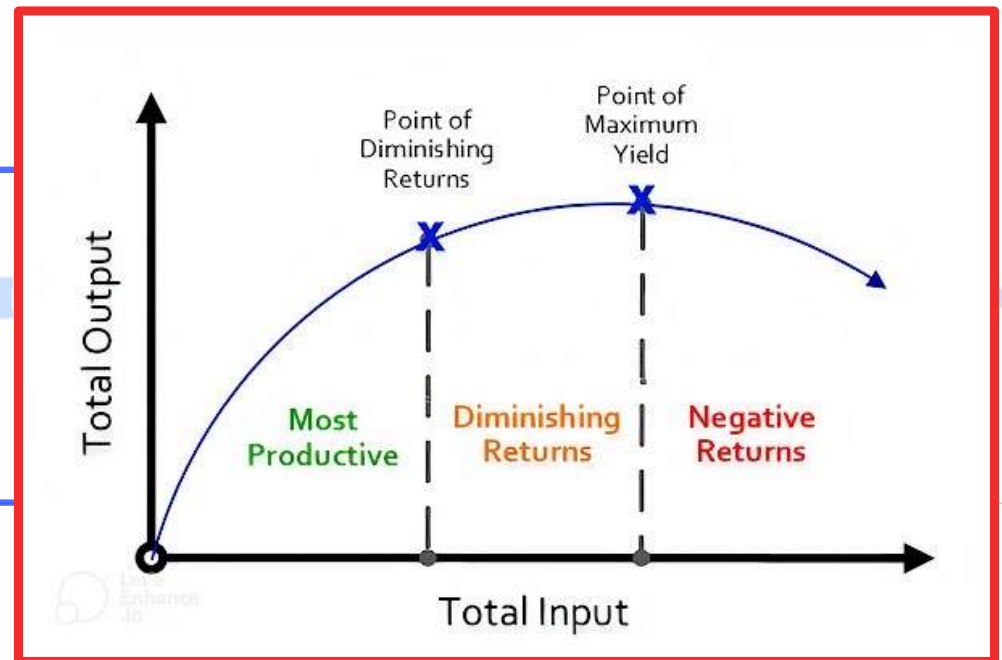


Diagnostic
CT / MR



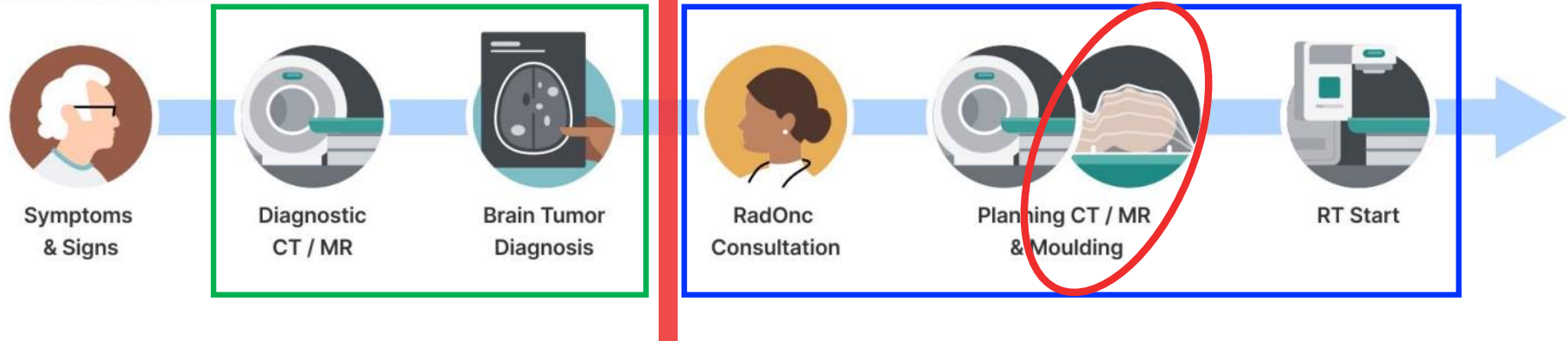
Brain Tumor
Diagnosis

1-4 weeks



The Problem OPORTUNITY

Current Workflow

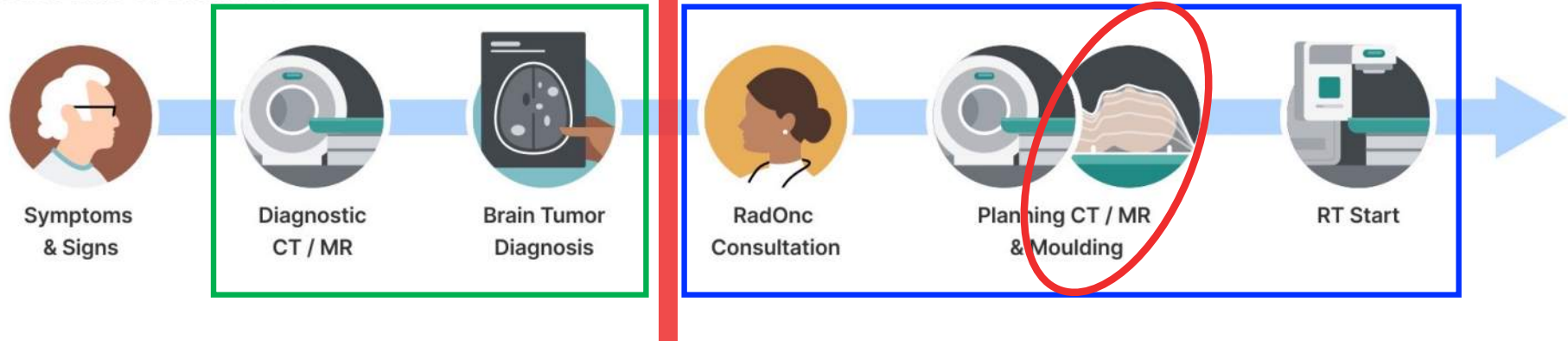


Interphase

*where answers/value
usually resides*

The Problem OPORTUNITY

Current Workflow

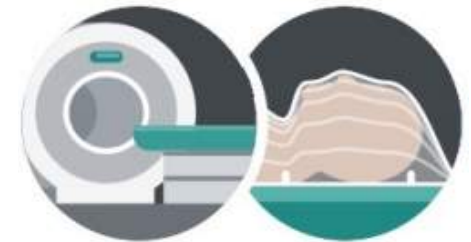


Interphase

*where answers/value
usually resides*

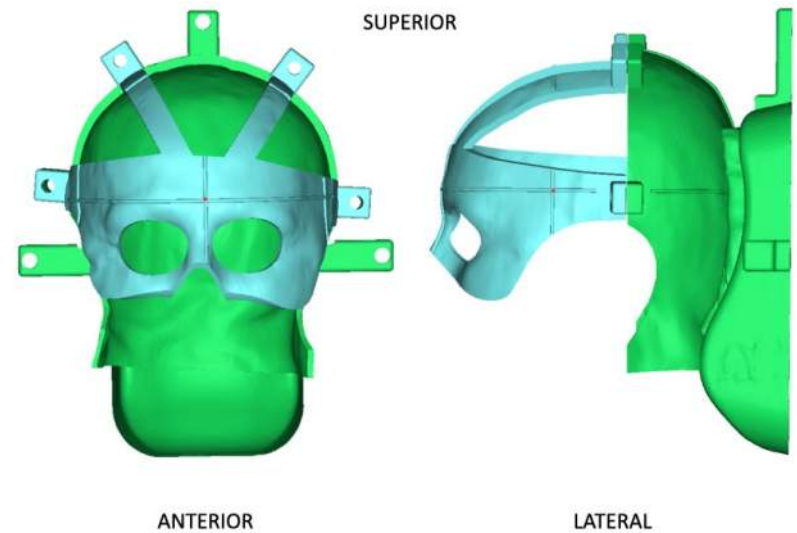
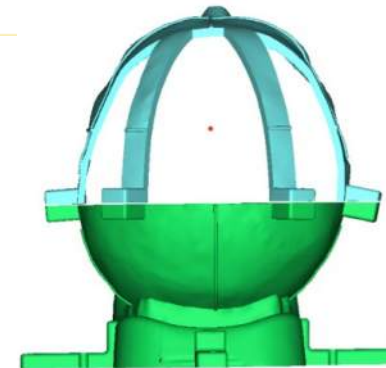
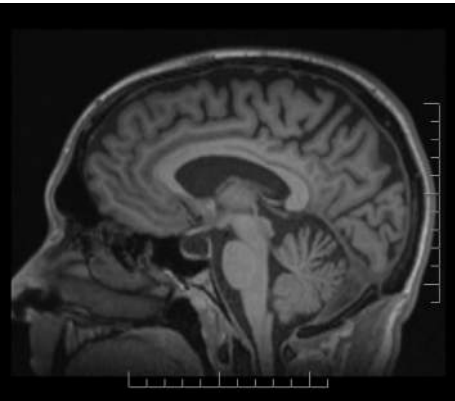
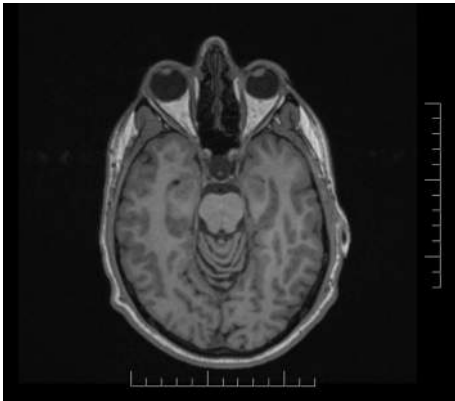
Questioning the Paradigm

- **CT Sim need?**
 - Most (all) patients already have imaging (CT, MR).
 - Electron density?: (s)CT
- **The “real” need:**
 - Reproducible positioning (?) and **immobilization**
- **The proposed solution:**
 - Use of diagnostic image-sets for **creating** a patient-specific immobilization device and **re-create** patient's random positioning during diagnostic MRI



Planning CT / MR
& Moulding

Whatever it takes!



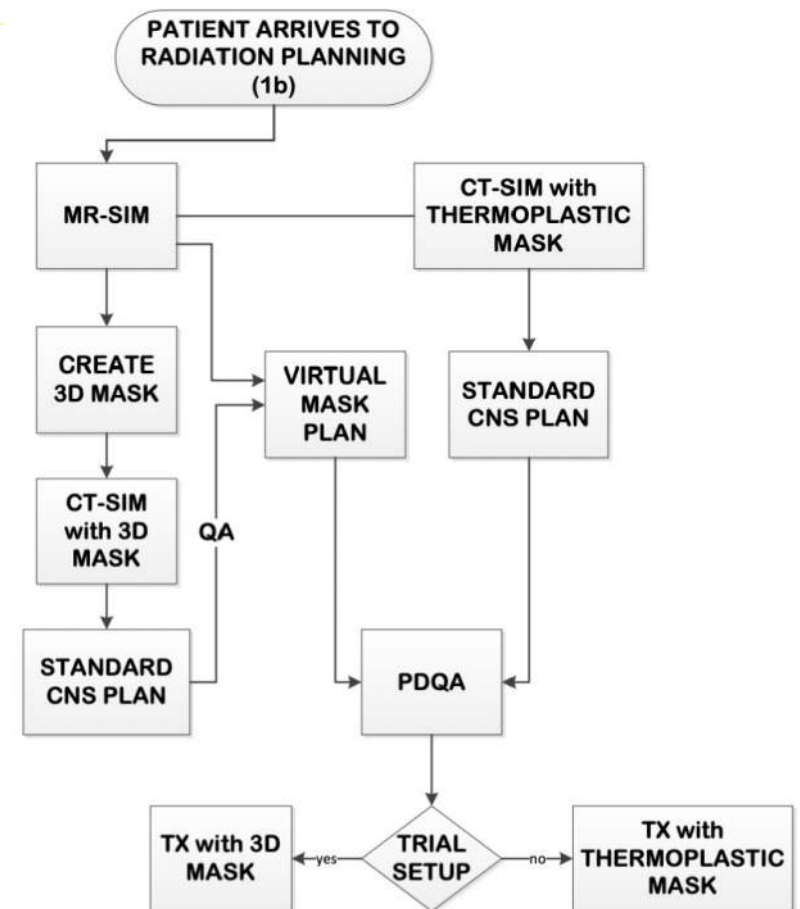
Study in a nutshell

40 patients
(20 each arm)

Study/Evaluation Future State
(safety redundancies)

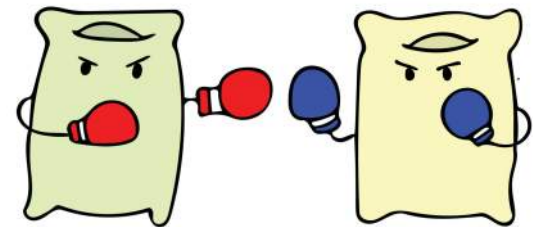
Largely unperturbed clinical
environment
(1 added CT-Sim and Tx Trial
setup)

Up/downstream unaltered



Results in a nutshell

- ✓ No 'fallback' cases
- ✓ Comparable reproducibility (i.e., inter fraction) and immobilization (i.e., intra fraction) performance
- ✓ No significant dosimetric impact
- ✓ No mask-related AEs
- ✓ Better patient-reported experience

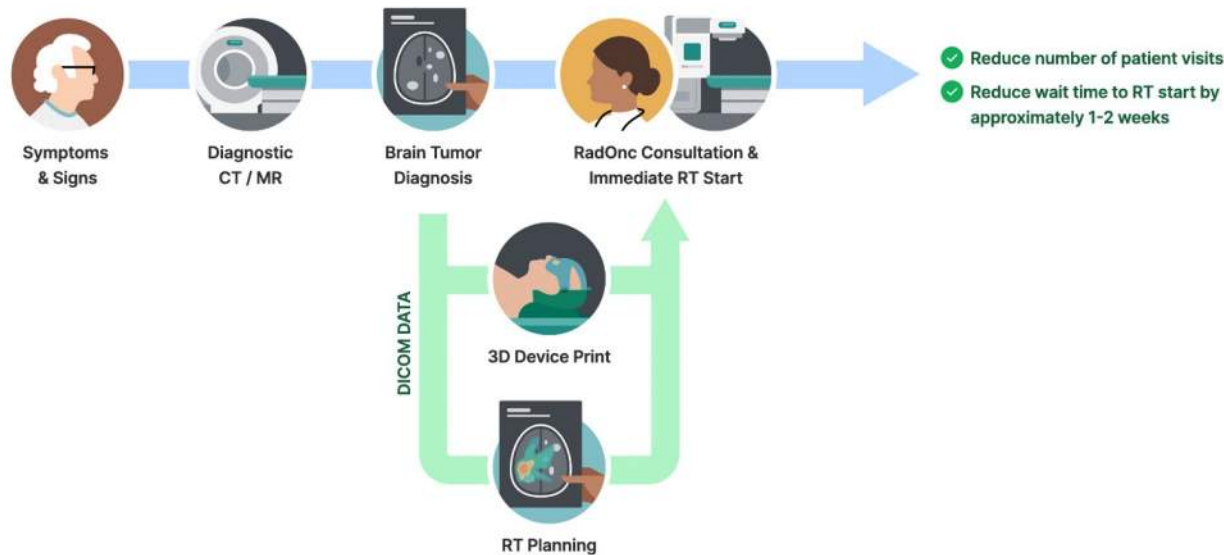


The REAL result...

Current Workflow



Proposed Workflow



1 vs 3 patient visits

RT start 1-2 weeks earlier

Avoid redundant imaging studies

Optimize use of resources

Increase throughput

Driver for MRI quality

Bringing it all together

A very important 'by-product' of all this
A very important opportunity to capitalize, *a priori*

Another (big!) Goodie

#whatchamacallit

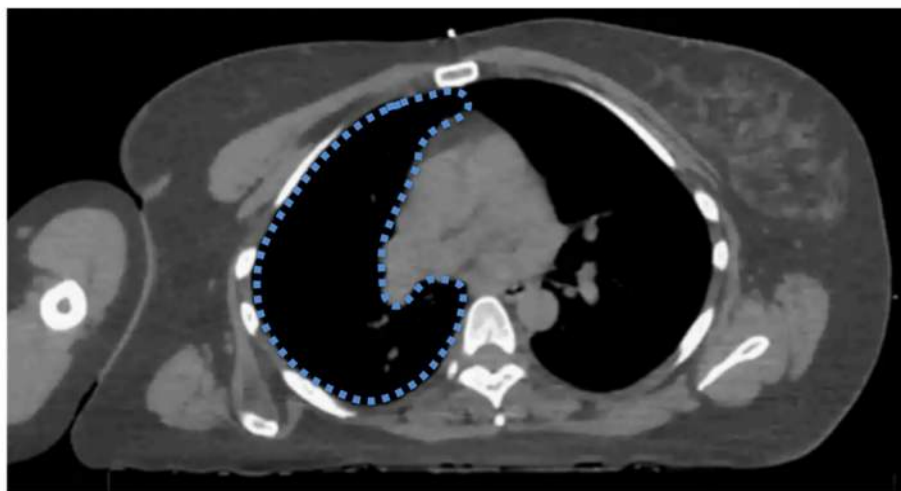


Another (big!) Goodie: Automation propels Standardization



- AGLET
 - keeps the fibers of the lace from unraveling
 - firmness and narrow profile that makes easier: to hold, to feed through eyelets
- Rumpelstiltskin Principle
 - **Naming** something **gives you power** over it: vocabulary allows you to think/talk about it.

Not only AI.... we're all about Big Data!, RWE!, etc



left lung

left_lung

Left Lung

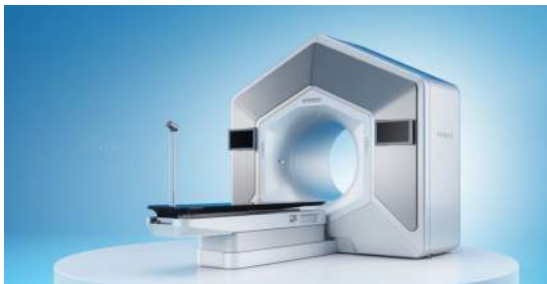
LT Lung

LLUNG

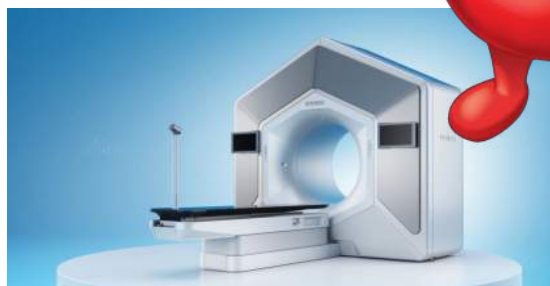
LEFT LUNG

Left_Lung

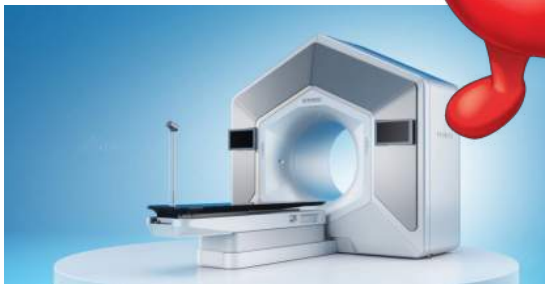
New Tech on the Block: adopt, adapt, and/or test (hypothesis driven)



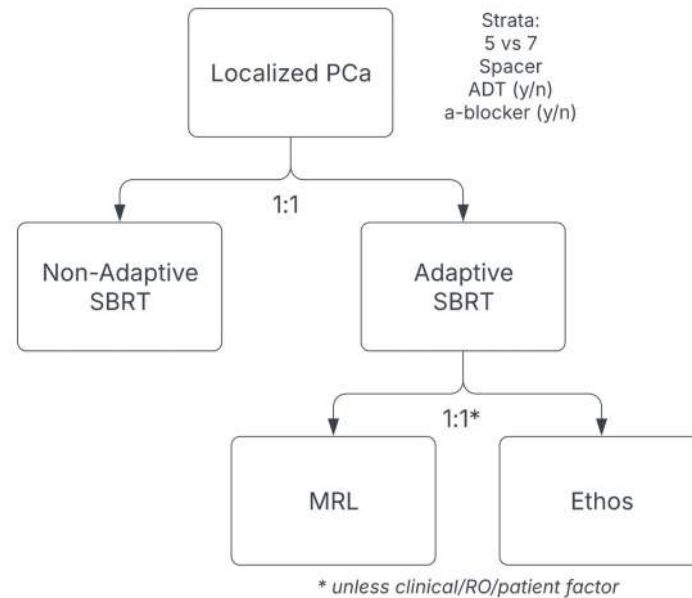
New Tech on the Block: adopt, adapt, and/or test (hypothesis driven)



New Tech on the Block: adopt, adapt, and/or test (hypothesis driven)



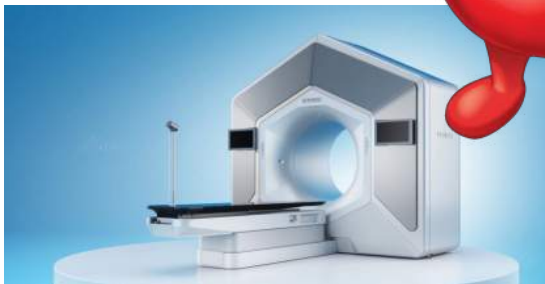
ASPIRE – is **A**daptive **SBRT** for **P**rostate vs Image-guided **R**adiotherapy a true **E**volution



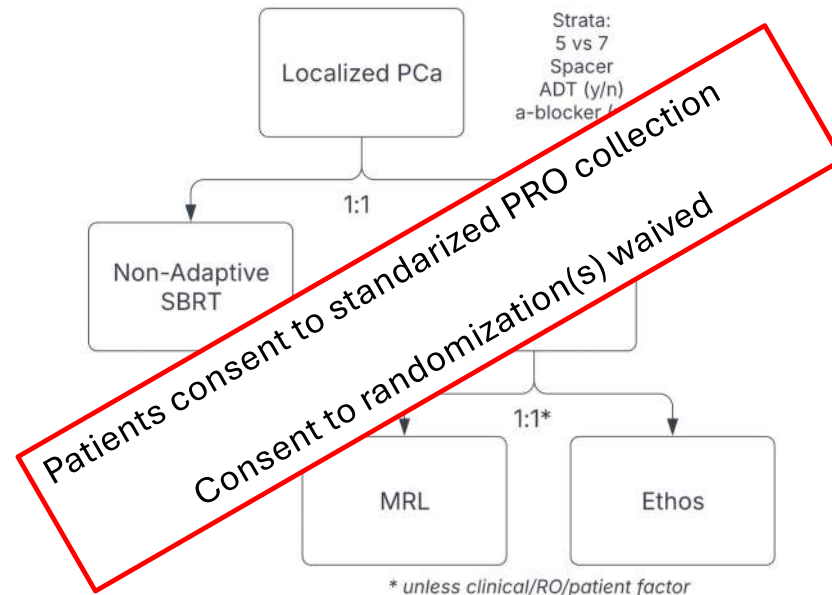
n=320

1ry endpoint: PRO QoL (GU) within 3mo

New Tech on the Block: adopt, adapt, and/or test (hypothesis driven)



ASPIRE – is **A**daptive **S**BRT for **P**rostate vs Image-guided **R**adiotherapy a true **E**volution



n=320

1ry endpoint: PRO QoL (GU) within 3mo

(some) Conclusions

- **AI is in the clinic** (or ready to be adapted, adopted, deployed)
- **AI is a tool.** Still need **skillset(s) + mindset** to maximize its potential.
- Will **enhance (hopefully transform)** way we work
- If you think it will replace you... it will. Alternatively, you can **think how to use it for working smarter** (i.e., human judgement is still irreplaceable)
- **Never say never...** Prediction algorithms = predictable, standarizable, etc. Humans (working with or without machines) #not.

(some) Conclusions

- AI is in the clinic (ed, deployed)
- AI is a tool. Still need to maximize its potential.
- Will enhance (hope to work)
- If you think it will replace you, you can think how the environment is still
- Never say never... (stable, but machines) #not. standarizable, etc.

